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A sustainability-focused innovation management and product development framework - Based on current practices at Mips

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En hållbarhetsfokuserad innovationsledning- och produktutvecklingsramverk - Baserat på nuvarande arbetsmetoder hos Mips

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Sammanfattning

Mips är ett växande ingrediensvarumärke som utvecklar säkerhetssystem för hjälmar. Deras kontinuerliga tillväxt har lett till behovet av att definiera innovationslednings- och produktutvecklingsprocesser. Mips har ett stort fokus på att arbeta hållbart och därav har deras forsknings- och utvecklingsavdelning bett författarna av denna uppsats att utveckla ett ramverk för att lägga till hållbarhetsaspekter till den nya processen.

Design Thinking metoden har tillämpats på arbetet och därför var tillvägagångsättet av arbetet som en produktutvecklingsprocess. Under *Empathize-fasen* samlades information angående nuvarande tillämpningar och därefter definierades behoven och kraven under *Define-fasen.* Därefter togs ett koncept fram med hjälp av de tidigare satta kraven och relevant litteratur.

Resultatet av arbetet är ett ramverk som lägger till hållbarhetsaspekter i den nya innovationsledning och produktvecklingsprocessen. Vidare så togs flera mallar fram som ett förslag för hur arbetsdokumenten skulle kunna se ut ifall Mips skulle vilja tillämpa ramverket i deras nya process. Några mallar har redan blivit tillämpade av Mips under arbetets gång.

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Abstract

Mips is a growing, ingredient brand that develops security systems for helmets. Their continuous growth has led to the need to define the Innovation Management and Product Development processes. Mips sustainability work is also being prioritized and therefore, the R&D department has assigned the authors of this thesis with developing a framework that would add sustainability aspects to the new process.

The *Design Thinking* method has been applied to this thesis and therefore, the thesis work was approached as a product development process. The *Empathise* phase was done first to gather information on current practices, and later define their needs and requirements during the *Define* phase. The *Ideate* phase followed where the concepts were iterated and developed based on the requirements previously defined as well as literature.

The result of this thesis is a framework that adds sustainability aspects as steps to be added to their current Innovation Management and Product Development process. Further, several templates were developed as a suggestion on how working documents could look like would Mips choose to apply our framework to their new process. Some of these templates were applied by Mips already during the course of the thesis.

FOREWORD

In this chapter, the foreword is presented.

Firstly, we would like to thank Mips for their active collaboration throughout this thesis. We especially would like to thank our Industrial Supervisor for the continued help and engagement, as well as everyone in Mips who contributed with their invaluable input during interviews and discussions. A big thank you to the whole R&D department for allowing us to conduct our thesis work and for their openness and cooperation during this project.

Secondly, we would like to thank Johan Arekrans, our supervisor at KTH, for supporting and guiding us during this time. Your input has provided invaluable help and your continued support has contributed to the successful outcome of this thesis.

Nathapong Khumhongsa
Deva Pérez Orviz

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NOMENCLATURE

In this chapter, the abbreviations used during this thesis will be presented.

Abbreviations

CAD	Computer Aided Design
CFO	Chief Financial Officer
COGS	Cost of Goods Sold
DP	Development Process
FoT	First out of Tool
GHG	Greenhouse gases
IM	Innovation Manager
IMP	Innovation Management Process
IP	Intellectual Property
KTH	Kungliga Tekniska Högskolan (Royal Institute of Technology)
LCA	Life Cycle Analysis
PC	Polycarbonate
PM	Product Manager
PMP	Product Management Process
R&D	Research and Development
RPD	Research / Pre-Development
RPDP	Research / Pre-development Process
SBTi	Science Based Targets initiative
TG	Toll Gate
UX-Design	User experience Design
VTL	Virtual Testing Lab

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1. INTRODUCTION

This chapter describes the background, the purpose and the limitations used of this thesis.

1.1 Background

It is widely known that since the 1800s human actions have had a massive impact on earth's long-term shifts in temperature and weather patterns, also known as Climate Change, caused by burning fossil fuels such as coal, oil and gas (United Nations, 2023). The process of burning fossil fuels generates greenhouse gases such as carbon dioxide and methane. These gases act as a barrier that keeps the sun's heat inside the atmosphere, rising the earth's temperature. To combat the negative impacts of climate change, the United Nations signed the Paris Agreement which aims to substantially reduce the global greenhouse gas emissions in order to limit the rising of the global temperature to 2 degrees Celsius during this century (UNFCCC, 2023).

To successfully prevent earth's temperature from raising, every contribution to the cause counts and the private sector has key role to play (Scheyvens et al., 2016). Therefore, the assumption that the main responsibility of management is to maximize the profit of the firm (Friedman, 1970) needs to be revised with more stakeholders in mind. To align with the Sustainable Development Goals (United Nations, 2015) the goal of business needs to consider the environmental, social and economic aspects of the organization's operations (Busch et al., 2018).

Sustainable development includes much more than the environmental benefits. It could lead to fostering more innovation in the organisation brought on by the development of products and services that meet sustainability requirements. Further, organisations that is sustainability oriented are also more likely to survive during crises, experience less share price volatility, and in the long run generate more revenues than those that lack a sustainability focus (Ortiz-de-Mandojana and Bansal, 2016).

Mips is one of the companies that has committed to the Paris Agreement by signing up for the Science Based Targets initiative (SBTi), which is a movement for the private sector to take urgent actions to combat the climate changes. Mips develops security systems for helmet companies to implement in their helmets. Their products are adapted to every new helmet model that is released and tested in-house to ensure maximum safety. Mips is an ingredient brand, which means that they are only responsible for a small part of the final consumer product, their security systems (Mips, 2023). Further, Mips business model is to sell the license of their products to the helmet brands. Moreover, the company only designs the products, that are later manufactured by a third-party. Mips has since 2020 implemented sustainability goals in order to contribute and take responsibility for the climate challenges mentioned above. As shown on the Annual Report 2021, Mips is a growing business (Mips, 2022), and therefore, their Research and Development (R&D) department has decided that they needed to map and define an innovation management and product development process to standardize the way ideas and projects are selected, evaluated and how they are turned into new products. This presents a valuable opportunity for our master thesis to study, define and create a framework that integrates sustainability aspects in the new innovation

management and development process since this is something that they have not yet developed or implemented.

1.2 Purpose

The purpose of this thesis is to help Mips achieve their sustainability goals, more specifically to help the R&D department to include sustainability in their processes. This will be done by researching current practices and designing a framework composed of steps and tools regarding sustainability. This framework will be applied to the newly mapped innovation management and product development processes at the R&D department at Mips. To achieve this, we will:

1. Conduct a qualitative study on current practices within the R&D department, as well as the newly developed innovation management and product development processes.

2. Suggest a framework for sustainability to be used in conjunction with the new innovation management and product development processes.

1.3 Relevance

The chosen topic is relevant for growing companies in the product development industry who are starting to standardize their product development processes to allow for the growth of the company while also considering their environmental impact.

For Mips this topic is especially relevant since management at the R&D department is currently mapping and defining an innovation management and product development process in the form of a framework. Mips has previously had a product development process, but it has however not been used by all the employees. Their current process development work will not extend to sustainability aspects. Therefore, this thesis will provide great value with the help of an in-depth study of their current practices, needs, and workflow, as well as the development of a framework to be added to their new process that will provide steps and tools regarding sustainability.

Academically, this thesis will contribute by applying theory regarding sustainable practices in growing companies to create a framework.

1.4 Delimitations

The focus of this thesis will only be on the environmental aspects of Mips, especially on what the R&D department can affect with their product development process. The reason for this is that Mips as an organization already excels at the social and economic aspects of sustainability. Further, this thesis has a time limit of 20 weeks therefore other sustainability in other departments will not be considered.

2 METHOD

In this chapter, the overarching method, and its phases, as well as the individual methods used in every phase will be presented together with a chronological description of the steps taken during this product development process followed to create the final framework.

The overarching method that was used during the thesis was the *Design Thinking* method according to M Luchs et. al., (2015). The method focuses on the user's needs and ensures that the product created is tailored to the user's needs. This was a necessary since the goal was to make a framework that would best cater to the needs and workflow of the R&D department at Mips. The different phases of the method also contributed to a clear and achievable structure of the thesis. The project was therefore divided in five phases, as defined by the author: *Empathise*, *Define*, *Ideate*, *Prototype* and *Test*. A Summary of the method used is shown in Figure 1.

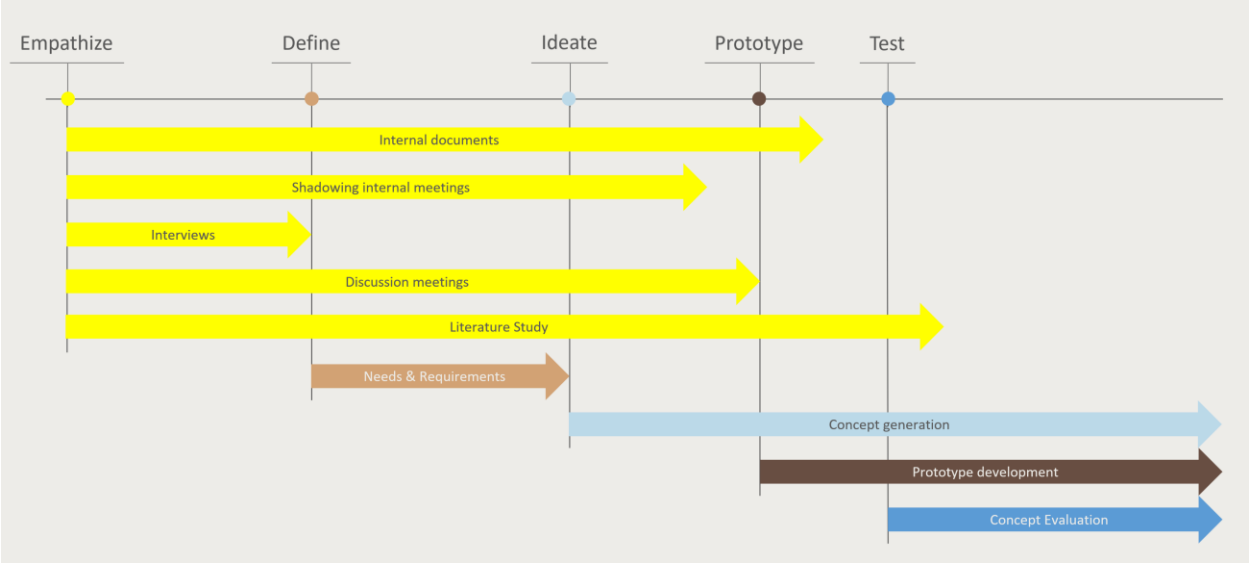


Figure 1: Summary of the method used for this thesis.

2.1 Empathise

During the *Empathise* phase a qualitative information gathering was conducted with a literature and a few other methods.

2.1.1 Interviews

The *Interview* method was chosen as an information gathering method because it produces general information on current practices, opinions, ideas, and issues (Wikberg Nilsson et al., 2015). The interviewees were encouraged to speak their mind and ideas. Semi-structured interviews and follow-up open discussions were conducted within a 30-minute meeting scheduled in a private conference room. Two interviews were conducted via Teams while

the remaining twelve were conducted in person. The interviews were recorded and later transcribed and analysed.

The interviewees were from the Research & Development (R&D) department at Mips, members of the Sustainability Group, and the Science Based Target Initiative (SBTi) subgroup. More specifically, at least one person with the following title from R&D were interviewed:

- Head of Product Development
- Senior Innovation and Product Developer
- Product Development Manager
- Product Development Engineer
- Technical Engineer
- Industrial Design Manager
- Project Manager
- Textile Specialist
- Polymer Specialist

Within the SBTi subgroup, all members were interviewed to get a good understanding of how Mips worked with sustainability as well as challenges, problems and needs that might be related to how the R&D department worked. The group had a representative from all the different departments within Mips: Operations, R&D, Implementation, Marketing, and Management. The Chief Financial Officer (CFO) and responsible for the overarching sustainability group was also interviewed. The roles of the interviewees were.

- Sourcing Manager - Operations
- CFO - Finance
- PR Director - Marketing
- Senior Technical Engineer – Implementation
- Senior Innovation and Product Developer – R&D
- Office and Resource Manager

For the first round of interviews a semi-structured interview guide was developed. A general guide was developed for the R&D team in general, adding some extra questions for the material specialists and for the process leaders that were involved in developing the new *Development Processes*. The guide (see Appendix A) was designed to understand the personal views on sustainability, current practices and opinions, as well as possible needs and requirements for the new *Development Processes*. Specific SBTi questions were added for members of the SBTi group. Only the relevant and more general questions were selected to be asked to the CFO. The questions in these guides focused partly on personal views on sustainability but the focus was the sustainability goals, current ways of working and current issues encountered when working with sustainability (see Appendix A).

Follow-up interviews were scheduled with the Material Specialists, Office and Resource Manager, Product Manager, Industrial Design Manager and Project Manager to discuss

more in detail current practices, processes, and their personal ways of working. For these interviews only follow-up questions were written and discussed.

The input from all the interviews was then synthesised and added to Miro for analysis where it was grouped in different categories to find patterns, common issues, and ideas for a possible solution.

2.1.2 Shadowing Internal Meetings

Shadowing is a method where a researcher or designer follows and observes the user in their natural environment (Interaction Design Foundation, 2020). The key aspect of shadowing for the researcher is to act only as an observer. Further, they are not to interfere with the participants as this might change the subject's behaviour in certain situations.

Several internal meetings were shadowed during the first months of the thesis in order to get a better understanding of how the R&D employees worked. The purpose was to find out non-spoken issues, possibilities, and opportunities for sustainability improvements. The same method was also applied to Mips' Sustainability Group, the focus of this group was to discuss ongoing sustainability projects at Mips. Their meetings were shadowed to discover if our framework could contribute or alleviate any frustrations or issues that are brought up during these meetings.

To avoid influencing the outcome of the meeting, said meetings were joined via Teams. Additionally, there were no interactions with any participants in the meetings and had their cameras turned off. Moreover, a point was made to document any frustrations with the ways of working that were noticed during the meetings. Furthermore, discussions that regarded sustainability and potential solutions for the thesis research questions were recorded in OneNote.

2.1.3 Discussion Meetings

During the time of the thesis Mips is testing their new *Development process* in a pilot project. At first the authors thought it would be beneficial to shadow these meetings as well. However, after some discussions with the Project leader and our supervisor at Mips, it was decided that an active participation would benefit both parties more. For Mips, it would be helpful to get feedback on the process while they were working with it. Any discussion brought up in these meetings would be kept in mind for future improvements. For the authors, it was useful to observe how the process is meant to be used and gain an understanding of where a sustainability framework could have the most impact in future projects.

2.1.4 Literature Study

A *Literature study* was conducted by first searching for relevant management theory regarding sustainability, innovation management, sustainable product development and similar areas. The papers were chosen by the number of citations and relevancy to the project. Additionally, two highly relevant papers were also recommended by the supervisor at KTH. Once a few good papers had been identified the *Snowballing* method was utilized

to find additional relevant literature. *Snowballing* is a method used to find articles of the same or similar kind of topics by utilizing the reference list or citations of the initial paper according to Wohlin (2014). This process was repeated iteratively on the newly found articles until enough relevant articles had been gathered.

To find the relevant literature a search was made in online libraries such as Emerald Insights, Sage Journals Online, ScienceDirect, SpringerLink, and Wiley Online Library. The keywords used was: Environmental New Product Development; Sustainable Product Development Process; Sustainable Innovation; Sustainability; and Environmental Specialist.

During the *Ideate* phase, an additional, in-depth study was conducted to find specific information on methods and options for the concepts.

2.1.5 Company Documents

Several company documents regarding sustainability goals and strategies, current processes within R&D, and other relevant information were read and analysed to gain knowledge, study, and compare how they are applied in practice. Additionally, documents regarding the new product development processes were also analysed to get a better understanding of them and to find out where and how our framework could be implemented.

2.2 Define

Before moving forward to the *Ideate* phase, all the information and input from the research phase (interviews, literature and internal documents and discussions) was compiled and summarized allowing the creation of a *Needs and requirements* list for the process. The list was composed of input from research, needs, requirements, and suggestions from interviews and discussions. The list was shared with the industrial and KTH supervisors for feedback and additional input. Once the supervisors had given their thoughts and feedback the *Needs and requirements* list was revised, and requirements were given a priority from 1 to 3. In this case number 1 meant high priority and given to requirements that had a high chance of affecting the final concept/framework. Number 2 were for requirements that could be good to have in the concept/framework as a good next step in Mips' sustainability work but were not crucial. Lastly, number 3 was meant for requirements that were deemed to have a smaller impact on Mips' sustainability work. Additionally, the rest of the inputs and needs were categorized as *Suggestions* (S) and as *Out of scope* (O). *Suggestions* were general sustainability recommendations for Mips to think about in the future and *Out of scope* was meant for issues that were out of this thesis' scope but part of the current practices.

In preparation for the *Ideate* phase, all the *Development Processes* were sketched in Miro to improve understanding and to facilitate the generation of ideas. The processes sketched were Innovation Management Process (IMP), Research / Pre-development Process RPDP, Product Management Process (PMP) and Development Process (DP). Since these processes were still in development during this thesis, the most current version was used during the *Ideate* phase with

knowledge that some aspects could change later. These changes were regularly checked and added to our sketches throughout the project.

2.3 Ideate

During the *Ideate* phase, several creative methods and tools were used to generate as many ideas as possible around the key sustainability strategies, issues, frustrations, and possibilities defined in the previous phase. A set of activities and tasks that conformed a sustainability framework, as well as examples of templates to be used, was added to the, still in development, new Development Processes.

2.3.1 Brainstorming

Brainstorming is a method for idea generation focused on quantity. Several rules must be followed: do not criticise, aim for crazy ideas, combine, and improve ideas and aim for quantity over quality (Nilsson et al, 2015).

The *Ideate* phase started with a short brainstorming where critical phases in the Development Processes where sustainability aspects were needed were identified. Secondly, a version of *Brainstorming* called *Brainwriting* was used (Nilsson et al, 2015). The authors generated as many ideas as possible for each critical phase in the Development Processes and repeated this throughout all four processes. Once both authors had generated as many ideas as possible for each key area of a single process, they moved to the second section of the method where the goal was to build on each other's ideas and further develop them. Once the *Brainstormings* were concluded, additional suggestions and ideas collected from the interviews and discussions were added to the author's results. All the results from the *Brainstorming* phase were saved on Miro for documentation purposes.

2.3.2 Concept creation

After all the *Brainstorming* sessions, all the different ideas needed to be evaluated and the best ideas selected based on Mips' needs. The options that best matched these needs, according to the information gathered in the previous phase, were selected.

Since the framework included four processes and the sustainability aspects needed to be added cohesively to all processes, an analysis of the ideas was conducted in order to find a common strategy and red thread that could be applied to the whole framework.

The first version of the concept (Version 1) included new steps to be added to all processes and suggestions on templates to be used during these steps.

The second version of the concept (Version 2) included a first version of all the templates selected and an updated version of the sustainability steps to be added to the framework.

The final version of the concept, two new templates were added and Version 2's templates was revised and reimagined to make them more visually comprehensive. These updates were based on feedback from Mips and the authors own evaluation.

2.4 Prototype & Test

The phases *Prototype* and *Test* of the *Design Thinking* method are presented in this chapter. Unfortunately, the time constraints of this project did not allow for practical testing of the framework. Ideally, this framework would have to be tested in a real project and followed from idea to finalised project. Instead, a thorough evaluation by process leaders at Mips was conducted, feedback gathered, and improvements made.

After the first version of the concept was finished a sketched visualisation was developed and presented to all four process leaders individually. During these sessions the progress and template suggestions were presented and in-depth questions regarding process details that were still unclear were asked.

The process leaders gave positive and negative feedback on the concept and even came with suggestions on how to further develop it. They also chose what templates they needed to be developed for the next version of the concept.

The second version of the concept was presented in a meeting where all Process Leaders and the Head of Product Development were present. In this meeting the whole framework was presented as well as some research to back the results. All Process leaders had an opportunity to ask questions and voice their opinions and feedback. Further, an opportunity was given to give further feedback with an email sent after the presentation with all the material and a protocol summarizing the feedback taken.

Before the final presentation for KTH, a presentation was held for Mips. This presentation focused on the characteristics of the framework and templates relevant to the company. Feedback that could be implemented before the presentation with KTH was taken into consideration.

Additionally, the progress was presented to the Industrial Supervisor, once a week during the whole project. This allowed to make sure that progress was heading in the right direction and that both the authors and the company were on the same page regarding the framework and contents of the thesis.

3 RESULTS

In this chapter, the results of the Literature Study, Current practices at Mips and the Concept creation and evaluation will be presented.

3.1 Literature Study

In this subchapter the result of the literature study and the relevant practices that can be applied to Mips and our framework will be presented. The theory in this subchapter is divided into three overarching topics: process, people and tools. The *Process* chapter regards sustainability in product development, sustainability requirements and how to engage society in the sustainability agenda as accompany. The *People* chapter talks about how sustainable practices can be improved with the help of human resources, such as sustainability managers and coordinators, and supplier involvement. Finally, the chapter *Tools* touches upon interesting tools that can be used to develop sustainable products.

3.1.1 Process

Many sustainable product development researchers agree that sustainability aspects should be considered in the earliest stages of the product development process (Bovea and Pérez-Belis, 2012; Luttropp and Lagerstedt, 2006; Ny et al, 2008; Sherwin and Bhamra, 1999). How the sustainability demands and specifications are defined when the product is being planned will steer how the whole product development phase will unfold (Luttropp and Lagerstedt, 2006). Some argue that they should be considered as early as during the development of the project brief (Sherwin and Bhamra, 1999), others state that they must be part of the requirement specification systematically balancing them against traditional requirements (Bovea and Pérez-Belis, 2012). Non-sustainable versus sustainable requirements will always mean that factors such as shape or price might need to be traded based on the customer and market's needs (Pujari, 2006). Attempts to include sustainability in a late stage of the process when a product design has been already set is very difficult and unpractical since its characteristics are already fixed (Ny et al., 2008) and any changes will have a high cost. Therefore, a close and early collaboration on environmental issues between new product development and environmental expertise is needed (Pujari et al., 2004).

Another important aspect mentioned is the importance of engaging society and consumers in the sustainability agenda (Sherwin and Bhamra, 1999). It is key that there is a clear information flow from the company on the proposed sustainability agenda of the company and their products.

3.1.2 People

Another important aspect found in research was the importance of having an environmental manager in companies working towards sustainability. Firstly, it will help increase responsiveness regarding sustainability within the company (Pujari et al., 2004). During

the product development process, it might also help having someone working full time with sustainability questions. This person should be able to answer all questions regarding the company's sustainability vision and goals (Hallstedt et al., 2013). Ritzén (2000) states that an environmental coordinator affects and gives higher priority to environmental issues and that it should be combined with strong signals from managements such as goal setting. An environmental coordinator is proposed by Pujari (2006) to be involved in the assessment of sustainable aspects at all gates of the product development process. Sustainability should be audited and be a part of tollgates in processes and decisions should be made considering the environmental criteria (Ritzén, 2000). Additionally, an environmental specialist working collaboratively during the product development process could help implement sustainable guidelines, but they will be most effective in the early and late stages of the process according to Genç and Di Benedetto (2015). During a transition phase towards sustainable product development, internal knowledge is required, and experts and product developers will benefit from being closely organized (Ritzén, 2000).

Early supplier involvement has also been proved to improve environmentally sustainable new product development (Pujari, 2006). Petersen et al. (2005) found that input from a carefully selected supplier can facilitate decisions taken during the product development process, improving design, and better financial performance. This is only possible when a complete and careful analysis of the supplier and their capabilities is conducted before the start of the project (Petersen et al., 2005). The value that an environmentally friendly supplier is only possible to follow and control if the relationship between the supplier and the company is transparent (Holtström et al., 2019). Finding good suppliers is a very important decision, the company must have knowledge and information up front of the supplier's capabilities and whether their cultures are compatible (Petersen et al., 2005).

3.1.3 Tools

An environmental database could be a very useful tool to store environmental impact data on material and components (Pujari, 2006) that will later help with Life Cycle Analysis (LCA) and guide product developers when making decisions (Hallstedt et al., 2013). This database could be a way of building competence and sharing knowledge within the company in order to increase knowledge on sustainability (Hallstedt et al., 2013). Further down the line this database could be a part of an information system, or a *knowledge base* used to market the new products by making accurate claims on the sustainability of the product (Pujari, 2006). Sustainability experts need to possess the "know how" while product developers only need the "know why" of sustainability aspects and changes that need to be done to the product to achive the sustainability goals (Ritzén, 2000). In combination with the database, all this knowledge should be available for them when making decisions during the development process.

During the product development phase, it is crucial to have as part of the process the identification of key sustainability aspects for the different components (Ritzén, 2000). This could be done with the help of several tools, such as the *Ten Golden Rules of Eco-design*. These rules are generic and must be customized to the company and the type of

product that is in development (Luttropp and Lagerstedt, 2006). The general version of these rules as written by Luttropp and Lagerstedt are presented below.

1. Do not use toxic substances and utilize closed loops for necessary but toxic ones.
2. Minimize energy and resource consumption in the production phase and transport through improved housekeeping.
3. Use structural features and high-quality materials to minimize weight in products. if such choices do not interfere with necessary flexibility, impact strength or other functional priorities.
4. Minimize energy and resource consumption in the usage phase, especially for products with the most significant aspects in the usage phase.
5. Promote repair and upgrading, especially for system-dependent products. (e.g. cell phones, computers and CD players).
6. Promote long life, especially for products with significant environmental aspects outside of the usage phase.
7. Invest in better materials, surface treatments or structural arrangements to protect products from dirt, corrosion, and wear, thereby ensuring reduced maintenance and longer product life.
8. Prearrange upgrading, repair and recycling through access ability, labelling, modules, breaking points, and manuals.
9. Promote upgrading, repair and recycling by using few, simple, recycled, not blended materials and no alloys.
10. Use as few joining elements as possible and use screws, adhesives, welding, snap fits, geometric locking, etc. according to the life cycle scenario.

Finally, the benefits of using tools in development processes to improve sustainability are directly affected by the success in their implementation to achieve an efficient work procedure (Ritzén, 2000).

3.2 Current Practices at Mips

This chapter will present how Mips is currently working with sustainability, how the product development process has previously worked, and how the new innovation management and product development process that is still a work in progress is supposed to work. Therefore, this thesis will base its work on the information given at the time of information gathering and major changes to the new product development process will not be considered.

As mentioned earlier in the report, Mips is an ingredient brand and therefore it not as simple as developing a product and release for mass production. Each helmet and helmet size that the Mips solution is to be implemented in must undergo implementation phase conducted by the Implementation department where they release the tool for mass production for the specific helmet. Therefore, the result of the product development that is conducted by R&D results in a release of guidelines, templates, and similar things that is handed over to the Implementation department.

Furthermore, the pilot development projects, where Mips is to create a new product, are commonly in collaboration with helmet brands.

3.2.1 Mips and Sustainability

As leaders in their field, Mips aims to lead by example also when it comes to sustainability. Sustainability had been a part of the annual report in previous years but in 2020, Mips published for the first time their environmental sustainability goals (Mips, 2021). In this report, Mips expressed their ambition to have 100% recycled materials in packaging, reduce the CO_2 emissions for travel by 50% based on emissions in 2019 per employee, shift to green energy in their facilities in Sweden and China and gradually reduce hazardous chemicals in their products. Since then, a lot of work and focus has been put throughout the company on how to improve their environmental impact.

It is a given for Mips to respect the laws and regulations that is applied where the company operates. Mips therefore seeks the assistance of legal advisors when needed to ensure that they have a comprehension of the local laws and regulations that are relevant to their business operations (Mips, 2023). The aim is to ensure that Mips operates within the legal limits and have strategies in place for potential risks and liabilities. This in turn includes all local sustainability regulations.

In 2021 in order to understand their impact, an analysis of the total climate impact of Mips' value chain was conducted by a third party (Mips, 2022). The result showed that material and product choice were key aspects to improve to achieve a bigger improvement in environmental sustainability. During 2021 Mips' goal of only using recycled packaging was accomplished. They also found new suppliers that could provide polycarbonate (PC) from other manufacturing waste. Mips also carried out a cradle-to-gate LCA on four of their most-sold products (Mips, 2022). Mips has for a long time worked with virtual testing that would substitute much of the physical testing that was previously conducted. This has helped the company to become more environmentally friendly during the product development process by consuming less material and minimizing shipping. In November 2021 the Mips launched the Virtual Testing Lab (VTL) that is currently used to test their products in development, reducing the number of physical tests conducted. Further, all the work previously mentioned contributed to Mips being able to phase out three high-emissions products from their range in 2022 (Mips, 2023).

In 2022 Mips signed the Science Based Target initiative (SBTi) and committed to reduce their emissions by 42% by the end of 2030. The SBTi is measured in different scopes. Scope 1 represents direct emissions that are under Mips' control. Scope 2 includes indirect emissions for energy purchased from their own operations. Finally, Scope 3 relates to indirect emissions from Mips' value chain that are not controlled by them such as the production of Mips' products in their supply chain which accounts for 95% of their total emissions. The other 5% represents emissions from Scope 1 and 2, from transportation of samples and the refrigerant at the offices as shown in Figure 2 (Mips, 2022).

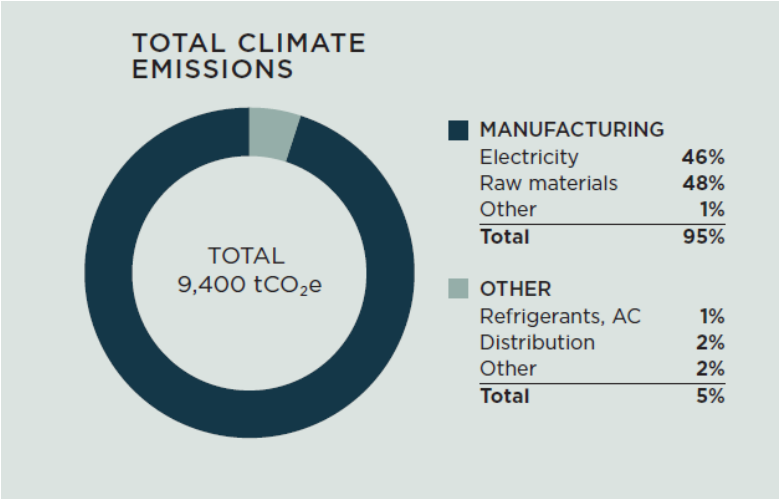


Figure 2. Mips' total climate emissions (Mips,2022).

During 2022 Mips identified the most sustainable manufacturing suppliers who invest in energy-efficient production equipment, solar panels, and renewable energy. The suppliers' improvement of their manufacturing resulted in a reduction of 118 ton CO_2e from 2021. Mips also continued mapping the environmental impact of two more products with lifecycle analyses. Mips also updated the product development process to include a climate calculation in order to be proactive regarding the impact of their products. In 2022 the company delivered eleven customer projects using post-industrial recycled plastic. By replacing polycarbonate, Mips most used raw material, with post-industrial plastic an average reduction of 7.41 $kgCO_2e/kg$ (Mips, 2023). Mips has also started to investigate after-life impact of their products and collaborates with Stena recycling to further explore the possibility of recycling some of their most popular products.

Sustainability in Mips is implemented and worked on across all departments (instead of sustainability being a centralized function). In 2021, Mips established a multifunctional sustainability group and three sub-groups who drive their sustainability efforts forward. In the main group ongoing projects are discussed as well as strategies on how to tackle the sustainability challenges in the future. The first subgroup is the SBTi group which focuses its efforts on projects and issues that will help achieve the SBTi goals before 2030. This group is currently led by the Sourcing Manager at the Operations department and has members from the Implementation and the R&D department as well as the Office and Resource Manager. The second sustainability sub-group is the circular product offer group which focuses on rolling out the new and more sustainable products to Mips' customers. The third sub-group is called the UN Global Impact where social sustainability efforts are focused such as conducting audits at the suppliers and factories to makes sure Mips Code of Conduct and human rights laws are being uphold. Mips is also currently recruiting a senior Sustainability Manger that will lead the sustainability initiatives at Mips.

A measure already taken regarding Mips' products is making sure that the lifespan of their products (the safety systems) is at least as long as the helmets' lifespan where they are implemented. They also always try to use recyclable plastic with as little impact on the environment as possible in their products. Mips' products are, when possible,

manufactured close if not in the same location as their costumers to minimise the environmental impact of transport.

3.2.2 Old Process

From the interviews, it was gathered that the R&D department in general had no strict way of working when developing a new product. However, the R&D department has successfully developed products over the years. The secret for these successful products is that employees has been encouraged to set time aside to explore and find new concepts. Further if the employees wished to start a project, they would have sought approval and keep management informed of its progress. This would allow for decision to proceed or halt the project. Since the department had limited manpower, it became common practice for employees to perform every part of the product development process. They would for example act as Product Managers and Product Designers simultaneously. Further, there are new Development processes from the early days of Mips for employees to follow. However, it was optional, and a few employees have expressed that it has been used at least 75% of the whole development process and others had their own way of working.

3.2.3 New process

A Project Manager, a Product Manager, a Product Development Engineer, a Senior Innovator, and a Product Developer collaborated to create the Development processes that are currently under construction. The developed method is based on a Phase-Gate or Waterfall Product Development, which is divided into four sub-processes. These subprocesses are designed to produce and select fresh candidates. The first and second sub-processes are not obligatory meaning a project can start from the third sub-process which the Product Management Process.

Before moving on to the Product Management Process (PMP) or the Research and Pre-Development Process (RPDP), a candidate is first chosen in the Idea Management Process (IMP). Candidates that need more research, such as those that might call for collaboration with other departments to be feasible or that simply call for a new production technique that Mips is unfamiliar with, are referred to the RPDP. Additionally, the candidates that aims to improve existing products can in some cases be sent directly to the PMP. In the RPDP the candidates are thoroughly investigated from all angles before being turned into concepts and delivered to the PMP for assessment to learn more about the possible markets, users, and business strategies. The concepts are then contrasted with one another, and the idea that best fits Mips' strategy and requirements is chosen to be turned into a product. In cases when the idea is not fully developed the Product Manager is able to send it back to RPDP. The Project Preparation Process is where resources and budget are assigned to the project once all the requirements for the product have been established. The Development Process (DP) starts after resources have been allotted to it. The idea or concept goes through five distinct phases and six Toll Gates (TG) during the DP: *Pre-study, Concept, Design, Pilot Production*, and *Product Release*. Each of these phases has a gate with conditions that must be met for the project to move on to the following phase. The sixth and final step serves as an evaluation of the entire project to provide information on potential future improvements to the Development Process or the next iteration of the product that has been newly developed. See Figure 3 for an overview of the new process.

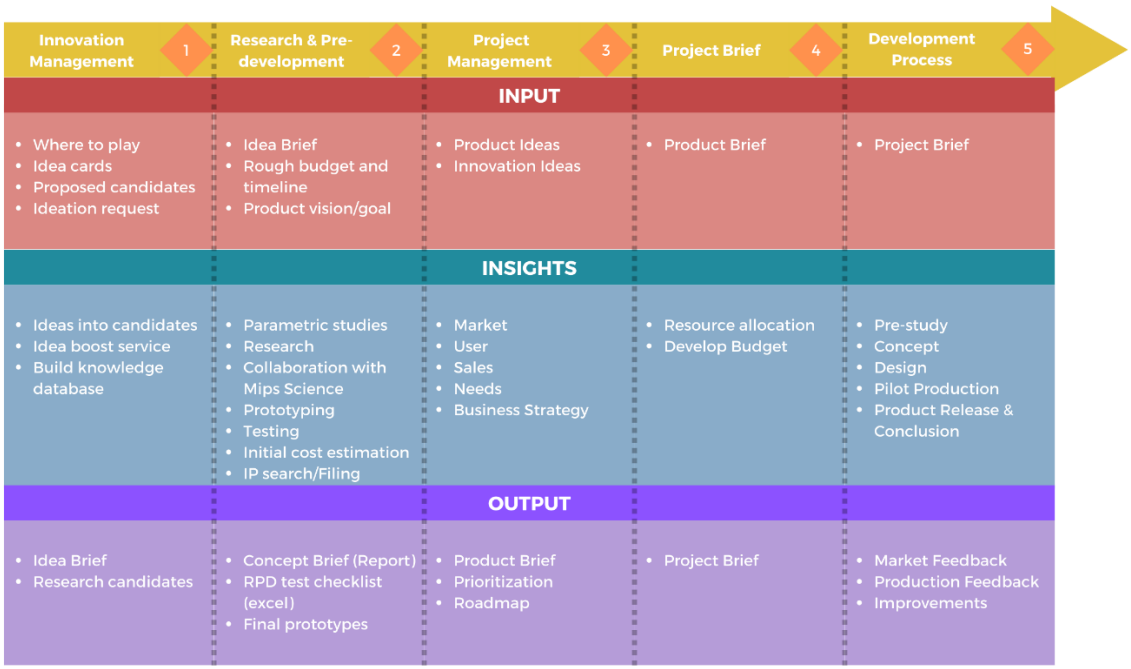


Figure 3. Overview of Mips' new Product Development Process.

Additionally, during the whole Development Process an Excel file, which will further be referred to as *Summary File*, is used to document key information and discoveries about the ongoing project. The document consists of multiple sheets that are intended to be updated continuously during the project and therefore it is used by everyone involved. However, the key contributors are the Innovation Manager, Research & Pre-developer, Product Manager, and Project Manager during their respective subprocess. The sheets in the document are as follows, *One-pager*, *Decision*, *R&D process checklist*, *Innovation handover*, *Pre-development test checklist*, *Product Brief Requirements*, *Product test checklist*, *Costs+BOM*, *and Product Specification.* The sheets relevant for the thesis will be explained in future chapters.

3.2.3.1 Innovation Management Process

The main objective of the Innovation Management Process or IMP is to prepare, select and deliver product candidates to the Product Management Process (PMP) or the Research Pre-Development Process RPDP. The IMP consists of two selection phases called first and second selection (see Figure 4).

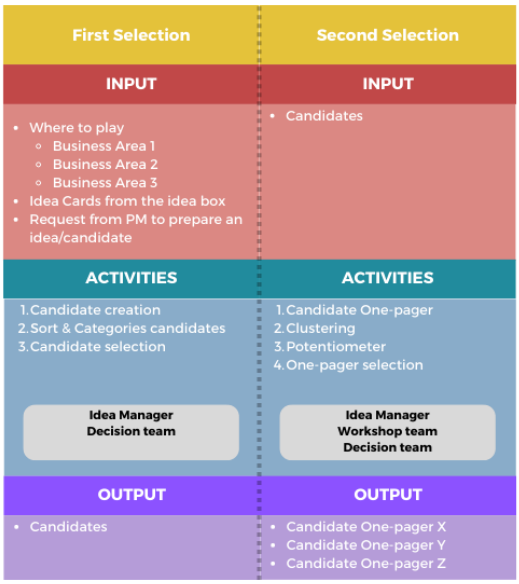


Figure 4. Overview of the Innovation Management Process.

The first selection consists of a phase called *Creation of candidates – baby status*, and it uses three inputs. The first input is a digital idea box where employees that are members of either Implementation, R&D, Science, and Operations have submitted their initial idea for a project or a product. The ideas are submitted as digital cards where the submitter can describe their ideas, what problem it might solve, and set product type and categories it belongs to. As of now the categories are as follows. *New product, Improvement* of an existing product, *Activity type*, and *Sustainability*. The second input are requests from the Product Manager (PM) or any other department at Mips. For example, employees in sales might get a lot of requests from helmet brands for a specific product that does not yet exist that can be worth looking into. The third and last input, *Where to play*, is tool to determine if Mips are developing products that aligns well with their long-term strategies and tactics. The activity for this phase is for the Innovation Manager (IM) to group and sorts all ideas from the three previously mentioned inputs. Sometimes different ideas need to be combined to make a more cohesive idea since some of the ideas at this point are quite vague. Once grouped and sorted, the ideas are then presented to a *Decision team* consisting of a Product Manager and Head of Product Development, together with the Innovation Manager they decide which ideas will move on to the Second selection which is the first selections output (see Figure 5).



Figure 5. Details of the First Selection.

In the second selection shown in Figure 6, the *Candidate One-pager* phase the Innovation Manager (IM) to creates candidates *One-pagers* where he or she sets the potential name, and vision for the candidates, write a simple description, set an expected budget, evaluate effort to sell, check patent opportunities, and unknown factors that might need further research. During the *Clustering* Phase, the IM clusters the candidates into product types, what activity it could belong to and which long-term strategy it belongs to. These product types and categories are the same as for the idea card. In the *Potentiometer* phase the candidates' chances of success, business potential, safety potential, and sustainability potential are evaluated by the IM. Additionally, the IM brings the candidate *One-pager*s to a workshop where they take the help of other Mips employees to Finalize the potential of said candidates. The *One-pagers* of the chosen candidates are then presented to a *Decision team* in the *Output* phase that decides if the candidates should be stopped, move directly to the PMP or if more research is needed.

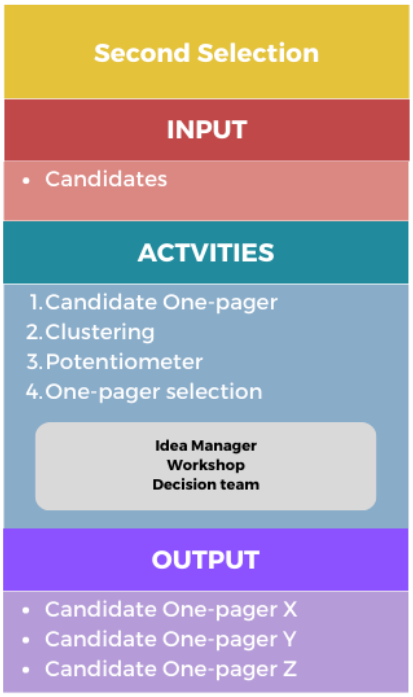


Figure 6. Visual representation of the Second Selection.

3.2.3.2 Research and Pre-Development

The Research and Pre-Development phase (RPDP) consists of four phases and TGs. The phases are as follows. *Initial Research, Pre-development Cycle, Concept Finalization, and Concept Handover.* See Appendix B for an overview of the RPDP.

In *Idea handover as shown in* Figure 7, the candidate *One-pager* is handed over to Research and Pre-Development from the IMP or PMP in a discussion meeting where the Innovation Manager, initial idea creator, RPD developer and/or Product Manager discuss the product's vision/goals and possibilities of the ideas. The outcome of this meeting is meant to decide whether the project should start, what personnel should be involved and set a rough timeline and budget.



Figure 7. Explanation of the *Idea handover* phase.

In the *Initial Research* phase an assessment is made, and the ideas' benefits, risks, and development areas are evaluated. Additionally, research regarding technology, materials, production methods, etc is made if necessary. Lastly, an Intellectual Property (IP) search or filing is performed (see Figure 8).

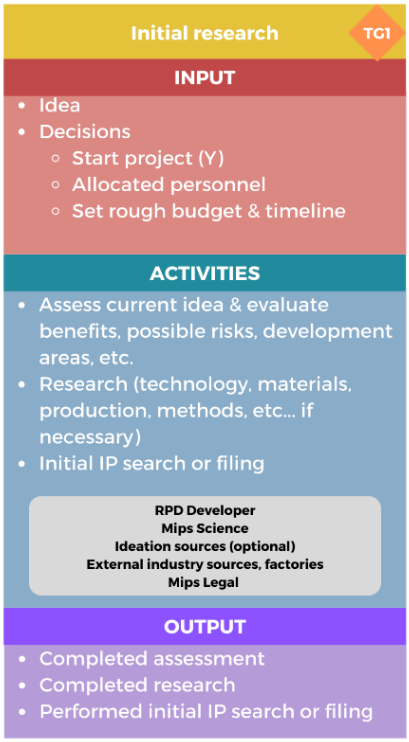


Figure 8. The *Initial research* phase.

In The *Pre-development Cycle*, an *R&D test plan* is created in collaboration with the R&D team in China. Further, design modifications are also to be made as sketches and CAD (Computer Aided Design) models. A prototype is also to be developed in this phase, preferably in-house prototyping, otherwise it will be outsourced. The prototype is then to be used to test functions, and parameter studies, and an assessment for improvements is to be made from the test results (see Figure 9).

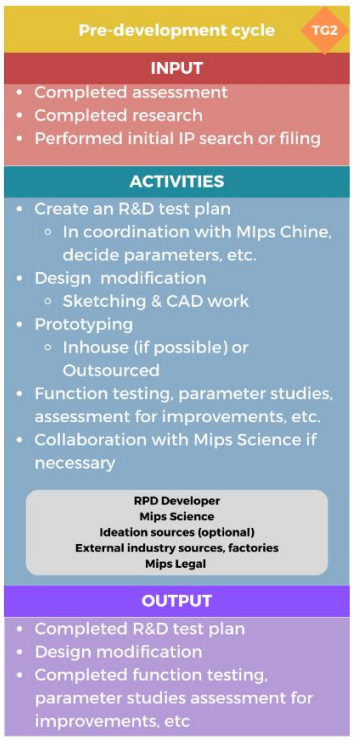


Figure 9. Contents of the *Pre-development cycle*.

In *Concept Finalization*, as shown in Figure 10, a final documentation is made with an initial concept specification including material and application recommendations, initial cost estimates, design ideas, etc. A complete RPD test checklist is also to be completed. Lastly, a final IP check is made.



Figure 10. Visual explanation of the *Concept finalization* phase.

During the *Concept Handover* phase, see Figure 11, the concept is handed over to the Product Management Process in a meeting where the final documentation, RPD test checklist, and the final prototype are presented. The meeting is also meant to assess the concepts' pros, cons, risks, and possible product goals.

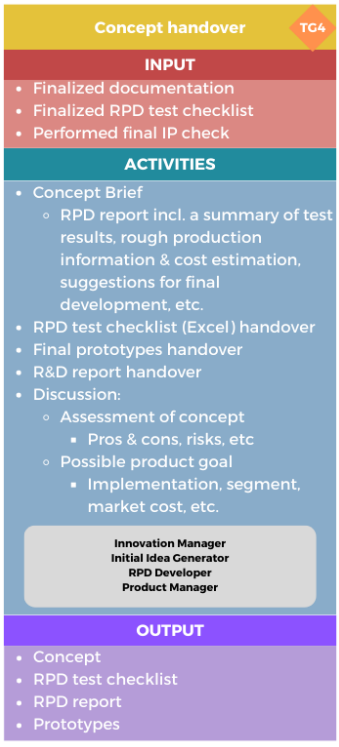


Figure 11. The *Concept handover* phase.

3.2.3.3 Product Management Process

During the *Insight* phase (see Figure 12, Figure 13, and Figure 14), the Product Manager (PM) evaluates different concepts against each other with help of a set of tools and a product board. The PM uses the tools and product board to gain an insight into the market,

the customers' needs and willingness to pay, estimate sales potential and possible business model is explored and set. The product board consists of the PM, Product Brand Manager, Category Moto Manager, Category Safety Manager, Head of Implementation and Head of Operations. As of now the PM represents the R&D department but this might change in the near future. After the evaluation the PM chose which concept to move forward with and the concept is productized by setting technical and standard testing requirements. These requirements are a mix of Mips own requirements and requirements that needs to be fulfilled to earn certain certifications. Once the concept has been productized the PM prepares a *Product brief* and develops a *Product Roadmap*.



Figure 12. The *Insights* phase.

Product board
Product Manager (R&D)
Product Brand Manager
Category Manager Safety
Category Manager Moto
Head of Implementation
Head of operations

Figure 13. The *Product board* and its members.

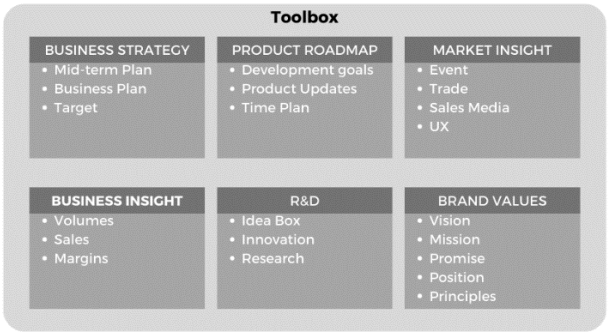


Figure 14. *Toolbox* and its contents.

3.2.3.4 Project preparation

In this phase the PM has a *Product brief* meeting with the other leaders at Mips and discusses how much resources the project need. Once everything has been discussed resources are allocated to the project in the form of time, development budget, and personnel.

3.2.3.5 Product development

The Development process consists of five phases and six decision gates. See Appendix C for an overview of the whole DP. The process starts with a decision gate called TG0, see Figure 15, and the main objective here is to decide if the Development process should start or not. For a project to start a project brief should have been released and the business concept/strategy been considered and agreed upon. Moreover, the decision to start the process is decided by the head of development, Product Manager, responsible developer, Head of Operations, Sourcing Manager, Product Marketing Manager, and Head of Implementation. Additionally, an Industrial Designer could be a part of the discussion if deemed relevant.

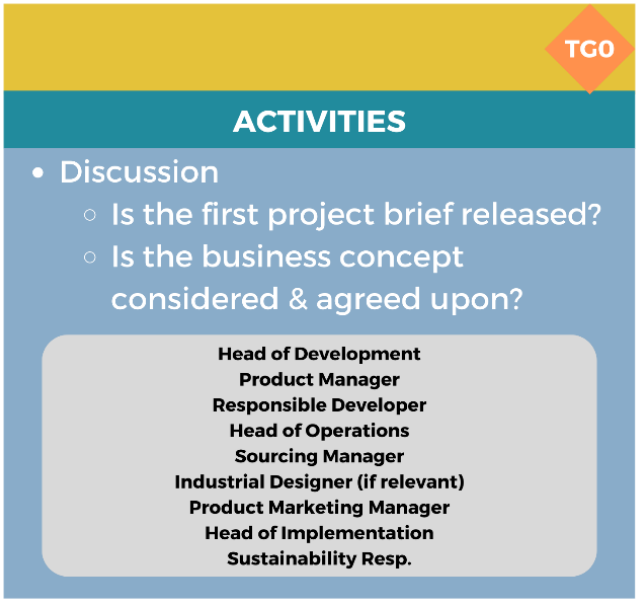


Figure 15. TG0, its activities, and participants.

The first phase of the Development process is called *Pre-study* and the requirements for it to start are an approved TG0, project brief, and product requirements. The requirements could for example be applicable standards, important functions, intended volumes, etc.

This phase aims to create a *Pre-study report* that includes suggestions for a technical solution, workload analysis, estimation of working hours, estimation of project cost, estimated Cost of Goods Sold or COGS (Operations), and check if requirement set can be fulfilled or updated. Moreover, a study to find suppliers for the product is also conducted at this phase. Further, the intended material for the product is evaluated and chosen at this phase by the product developer based on the requirements given to them. Once the *Pre-study report* is finished it is time for TG1 where the *Decision team* looks through the report and assess if the estimation and results are reasonable for the project to move on to the next phase otherwise more research needs to be done. For a visual explanation of the Pre-study phase see Figure 16.



Figure 16. Visual explanation for the *Pre-study* phase.

The second phase of the Development process is the *Concept* phase (see Figure 17) and the requirements for it to start are an approved TG1, a revised project brief, updated product requirements and an approved *Pre-study report*. The activities in this phase are to create a working concept with the chosen material, perform initial tests and verify the functions of the said concept. The kinds of tests to be performed are a fitting test and a characteristics test which includes product appearance, form, etc. As of now, there is no set method to evaluate the test results. Further, a *Design specification* to validate how the requirement are fulfilled is also created. Additional activities are to create a project plan that includes updating the COGS, confirming & setting material, deciding suppliers, and updating the time. After a completed concept phase, the process moves to TG2 where key stakeholders look over if the concept is functional, if it has the required functions, and if a design specification and a project plan have been created. If everything is in order the project moves on to the third phase otherwise get a No Go and the documentation needs to be revised or the put to a halt.

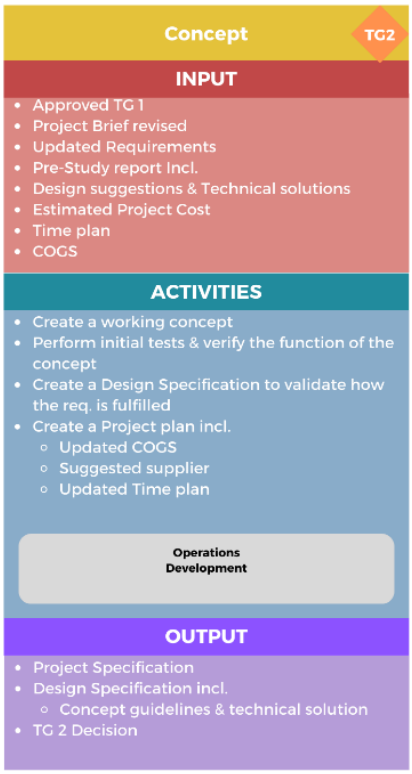


Figure 17. Input, activities, and output for the *Concept* phase.

The third phase is the *Design* phase (see Figure 18) and for it to start it needs an approved TG2, a released product brief and requirements, a design specification, and a project plan. The phase starts off with an iterative process to create, produce, and test a working prototype to find and address any issues with the product. The product must also be verified against the requirement specification and at the same time documented in a report. As of now there is no set way to evaluate these concepts but generally the product will be evaluated by fit both with test groups internally and externally. Further, things to evaluate are characteristics, appearance form etc. Once everything has been resolved and the concept is finalized the COGS is updated, and necessary tooling for serial production is ordered. Lastly, marketing material for internal release should be created and reviewed. The third gate, TG3, is to check and verify if the concept is working, if the requirements are correctly verified, and if the verification report is released and the result approved. Additionally, key stakeholders also look over if the updated COGS has been accepted, the marketing material is released for internal use, an internal release has been performed and the necessary tooling for serial production has been ordered.



Figure 18: The contents of the *Design* phase.

The fourth phase (see Figure 19) is called *Pilot Production* and the requirement for it to start is an approved TG3, finalized design of the product, a verified requirement/needs list and an approved First out of Tool (FoT). During this phase, the approved product is sent to be produced in the intended serial tools at the intended factories. The process then moves on to TG4 to evaluate the result of the pilot production is acceptable and meet the expectations. Further, is to check if the product has been verified correctly and if everything is in order for the product to move to the fifth and final phase.

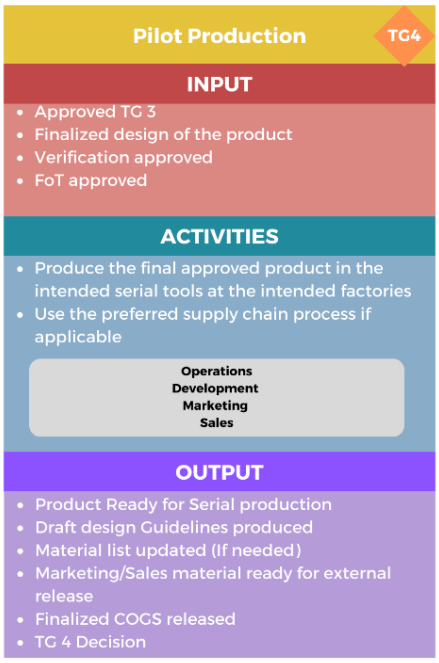


Figure 19. Overview of the *Pilot Production* phase

Once everything in TG 4 has been approved and the product is released for mass production the product moves to its last phase which is *Product Release & Conclusion* (see Figure 20). In this phase, the *Design guideline* is finalised and released, and the marketing is released. The *Design guideline* is a document to help Implementation to implement the newly developed product into the helmets and it is written by the product developers. Furthermore, meetings are held to share information with the stakeholders involved and a meeting is also held with everyone involved in the project to evaluate and discuss what has been learned during the project's runtime. The lessons learned will be used to improve or change the product Development process in future projects.



Figure 20. *Product release & Conclusion* phase.

Lastly, in the TG5 meeting (see Figure 21) the release of the final design guidelines and marketing/sales material are checked. Additionally, the gate is also meant to check if the information-sharing meetings and *Lessons learned* have been performed and documented. Once everything is in order the project concludes.



Figure 21. Final gate, TG5, for the Development process.

3.3 Needs and Requirements

Based on the input from the interviews, the literature study the *Needs and Requirements list* was formulated, see Appendix D for the complete list. In the list needs are translated into requirements and given priority one to three as described in Chapter 2.2.

The overarching theme of requirements given priority 1 is the continuous reminder of sustainability throughout the whole product Development process, from the initial idea all the way to the project's conclusions. This in turn led to including sustainability to some extent in all the subprocesses as, for example, guidelines, evaluation, or verification. The requirements given priority 1 state the continuous update of sustainable requirements set for the product. Meaning that sustainable requirements set in the early subprocess should be vague and as the project moves forward the requirements become more and more measurable.

Further, most of the requirements given priority 2 fall into the category of environmental impact of materials. The topic of these requirements are after-life implications and parts with high environmental impact such as high emissions, usage of toxic substances, etc.

Lastly, requirements given priority 3 cover cross-functional teams, international collaboration, and knowledge building. The solution for these requirements would be a database or platform that employees could use to store and access information gathered. This solution does however not match the goals of this thesis and therefore they are given priority 3.

3.4 Concept

In this chapter, the different concept versions, Version 1, Version 2, and Final Solution are presented and its relation to theory briefly discussed.

3.4.1 Version 1

In this chapter the features of Version 1 that are common to all the processes will be described. During the development of Version 1 all processes in the framework, presented in the New Process chapter, were analysed and the chosen ideas from the *Brainstorming* steps were added as green text (see Figure 22). These steps will be explained in detail in the coming sub-chapters. Further, as a natural effect of a phase-gate model is that anything added in a phase would naturally be a part of the next gate. For example, if a sustainability study would have been added to the *Pre-study* phase it would also be added as something that must be checked or verified in the following Toll Gate.

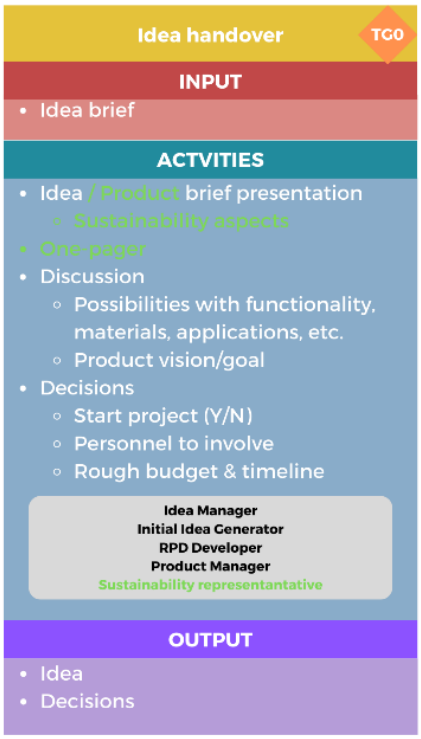


Figure 22. Example of how steps (in green) were added to the phases.

The red thread found after the *Brainstorming* and the *Empathise* phase was that sustainability needed to be present from the beginning in the processes and that it should be a part of the final report and Tollgate documentation as an output in every process (Luttropp and Lagerstedt, 2006) (Bovea and Pérez-Belis, 2012).

The first common characteristic of all processes is that the important decisions taken during the Toll gates of all processes are taken by a *Decision team*. The members of this team were not defined at the time, but it was deemed that a Sustainability representative that had sustainability interests as their priority, would be necessary based on recommendations from Pujari (2006) and Ritzén (2000). This person would need knowledge about Mips' sustainability goals and strategies, best practices within sustainability management and operational knowledge within Mips. During Tollgates and decision meetings, a protocol with decisions, reasoning, and attendance is recommended to be used to improve traceability.

The *Eco-design rules* (Luttropp and Lagerstedt, 2006) were adapted to Mips and rewritten as a key part of the development of Version 1. These *Eco-design rules* are to be used in all processes as a guide on important sustainability aspects to be taken into consideration. As Luttropp and Lagerstedt mention in their article, these *Eco-design rules* should be adapted to the kind of products that the company develops. The rules presented below are a proposition that the authors have defined based on the current and short-term vision of Mips learnt during the *Empathize* phase. The *Eco-design rules* are not listed in a priority order. See Table 1 for examples on how Mips could Implement the rules.

Table 1. Eco-design rules and examples for how Mips can implement them.

	Rule	**Examples of areas for implementation**
1	Reduce or eliminate the use of toxic substances in products	Trace and minimize chemicals in textile production and coatings / Study and minimize chemical impact from coating of parts
2	Minimise energy and resource consumption in the production phase and transport through minimise waste, use same parts when possible	Use the same parts for different applications / Design to maximize material efficiency during production (in collaboration with supplier)
3	Minimize weight and material usage in products, if such choices do not interfere with necessary functional priorities.	Minimize padding, part thickness, how much material can be removed before part is no longer functional
4	Consider impact of raw materials (origin, emissions, recyclable, recycled)	Study social sustainability, where does the material come from, is it virgin material? could re-purposed/ recycled material be chosen instead?
5	Minimize use of materials that are hard to recycle or re-purpose or that have a high environmental impact at any point of their life cycle (including end-of-life).	What part have highest impact? Can those materials be replaced? Considered impact during all life cycle?
6	Promote long life of products. In implementation, product life should be at least as long as helmet life. Timeless, inclusive design for emotional durability.	Quality materials and durable design.
7	Invest in better materials, surface treatments or structural arrangements to protect products from dirt, corrosion, and wear, thereby ensuring reduced maintenance and longer product life	Design for easy maintenance to lengthen product life

8	Prearrange recycling through access ability, labelling, modules, breaking points, and manuals.	Maintenance, end of life, recycle instructions / Mips module easy to remove for recycling
9	Promote recycling by using few, simple, recycled, not blended materials and no alloys	
10	Use as few joining elements as possible. Use tools that allow for easy disassembly such as snap fits, geometric locking, adhesives...	Avoid glue, plastic welding

In Version 1 four templates were chosen to be suggested for development to the process leaders during the first round of feedback sessions. These templates were. the *Sustainability Requirements list*, the *One-pager*, and the *Design log*.

Firstly, the need for sustainability requirements for all processes was evident during previous phases and therefore a *Sustainability Requirements list* was suggested as a possible template to be developed (Bovea and Pérez-Belis, 2012). The idea was that the list would evolve as the project passed throughout the processes. The requirements for the first processes IMP and RPDP should be vague to not hinder innovation, output requirements of the RPDP should be clearer, and the requirements from the PMP should be measurable and specific. The requirements would also be different for incremental innovation, where measurable LCA data would be available, than for radical innovation, where completely new products were created and therefore, it would be difficult to find specific data to compare.

Secondly, a *Design log* was suggested for development. The information gathered up to this point led us to believe that there was no standard method to report and document progress during the processes. The idea was to document design iterations, as well as test results in one file and build upon the concept during each process. One process leader had sometimes worked with such a file during the old process, but it was not a common practice. Further discussion with Mips, led to the discovery that test results were going to be stored in the *Summary file*. The project progress and decisions made would also be stored as a *Decision log* in this file. Therefore, the only information that needed to be stored in this template was the design iterations.

Finally, the last template the authors suggested to be used was the *One-pager* initiated during the IMP. The original document was created by the Innovation Manager at Mips. This document would be a part of a *Summary file* in Excel that would store important results and information during all processes in different sheets. In the original document, sustainability was only represented as a potential which was a subjective measure. In order to better represent and prioritize the sustainability needs, a row for sustainability was added by the authors where relevant sustainability information was to be filled at the end of every process (see Figure 23Figure 23).

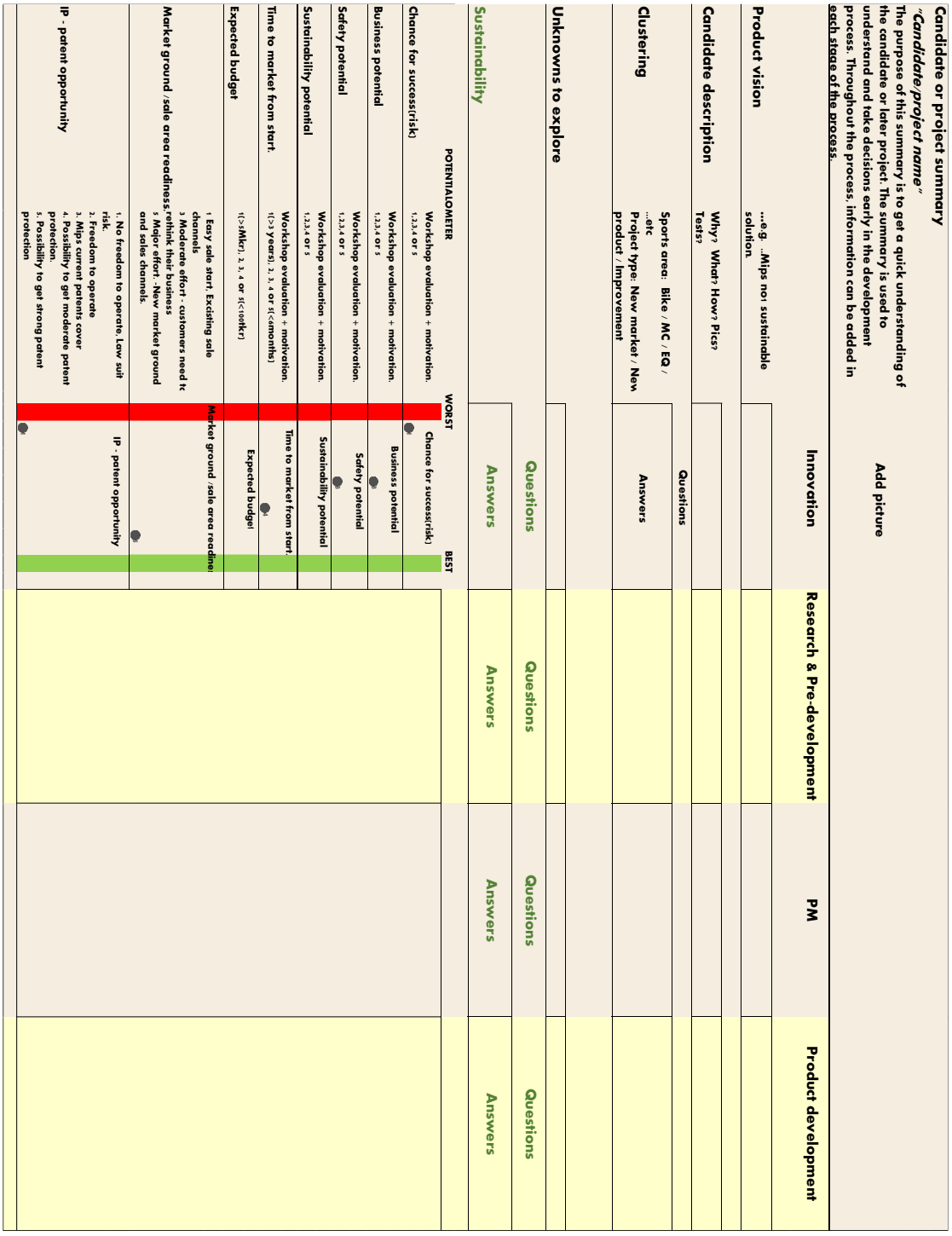


Figure 23. Version 1 of the One-pager with added row for sustainability questions and answers in green text.

3.4.1.1 Version 1. Idea Management process

According to Sherwin and Bhamra (1999), Pujari et al. (2004), and Ny et al. (2008), there is a general agreement that it is vital to incorporate sustainability as early as possible to develop a sustainable product therefore more sustainability topics has been suggested to be used when ideas are generated. The topics added is also meant to act as a constant reminder to Mips employees to think about environmental sustainability as early as in the initial idea and thus eliminating the excuse that "It is too late to make the product sustainable" once the product has gone through most of the DP. Additionally, this would also make it easier for the IM to sort and group ideas by having more topics/categories. Moreover, it will also

become easier for the IM to combine ideas because he/she would instantly know what categories and topics each idea falls into. See Figure 24 for suggested sustainability topics that ideas might fall into.

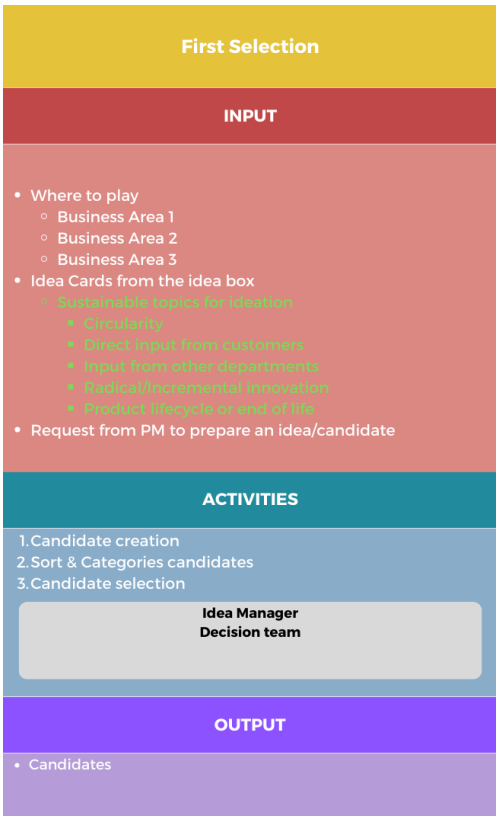


Figure 24. Sustainability topics added in green text.

3.4.1.2 Version 1. Research and Pre-Development process

The Research and Pre-Development process (RPDP), as presented in chapter 3.2.3.2 is the first process where the candidates are researched, and concepts developed. Therefore, it is important that sustainability is present from the beginning of the process (Pujari et al., 2004). See Appendix E for an overview of changes made in the Research and Pre-development process.

Already in Tollgate 0 (TG0) information about sustainability aspects was added to the contents of the brief as recommended by Sherwin and Bhamra (1999), see Figure 25. Since the brief can come directly from the IMP or the PMP, it will contain vaguer information or will be more developed, respectively. The *One-pager* as a summary document will also be an *Input* document present in all projects. The *Decision team* will be present during all gates and, as mentioned before, will include a sustainability representative (Ritzén, 2000).

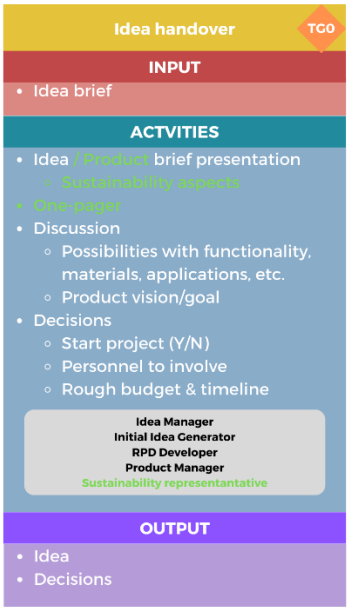


Figure 25. Version 1 of the *Idea handover* phase with sustainability aspects added in green text.

During the *Initial research* phase, the candidate is assessed. An assessment of sustainability with the help of the *Eco-design guidelines* unfold (Luttropp and Lagerstedt, 2006) was also added to this step. Later in this phase, a new task, the Sustainability research was added. In this research the following topics were suggested: emission, alternative materials, green production possibilities, eco-design, and after-life possibilities. The work conducted in this phase is to be finished by TG1. Changes made are shown in Figure 26.



Figure 26. Version 1 of the *Initial research* phase with added steps or tasks in green text.

In the *Concept finalization* phase suggestions for possible sustainability requirements for the next phases were added. At this point the *One-pager* and its sustainability potential are to be updated. These tasks are to be performed by TG3 (see Figure 27).



Figure 27. Version 1 of the *Concept finalization* phase with added steps in green text.

During the phases *Initial research*, *Pre-development cycle* and *Concept finalization* a continuous documentation of sustainability findings were added in a form of a *Design log* as suggested by Pujari (2006).

Added to the *Concept handover phase* was for the RDP developer to handover *Suggestions of sustainability requirements*, an updated One-pager and sustainability findings (see Figure 28). The sustainability findings will be included in the R&D report as an output document together with an updated *One-pager*.



Figure 28. Version 1 of the *Concept handover* phase.

3.4.1.3 Version 1. Product Management process

New products to develop are chosen during the Product Management process (PMP) and therefore, sustainability needs to be present during all phases of this process. There are two kinds of inputs for the *Insight* phase: concepts from the RPDP and candidates from the IMP. Therefore, the sustainability steps added to this phase are different. For candidates from the IMP only sustainability potential is required since the candidate is too young, and no specific research has been done. For the concepts that have undergone the RPDP a suggestion of sustainability requirements based on the sustainability research were added to the concept brief and *One-pager* as input (see Figure 29). A sustainability representative, as suggested by Hallstedt et al. (2013), was added to the Product board (see Figure 30). Some areas were also added to the topic *Areas to evaluate* as shown in Figure 29. Firstly, on the subtopic *d. Needs* the user's needs regarding sustainability were added. Further, on the subtopic *e. Business strategy*, a sustainability study of the business model was added to find out how the product would contribute to reaching the sustainability goals.



Figure 29. Version 1 of the *Insights* phase with points added in green.



Figure 30. Version 1 of the product board.

In the *Toolbox* used during this phase, several tools were added (see Figure 31). In *Market Insight* the customer's sustainability views are to be taken into consideration, in *Brand Values* and *Business Plan* sustainability were added as a factor to study, and finally a whole new tool was added called *Sustainability Roadmap*. This last tool should be aligned with the *Product Roadmap* so that a clear correlation is shown between the sustainability goals and milestones and to make sure that they are aligned with the products planed for release, as well as the products that need to be phased out.

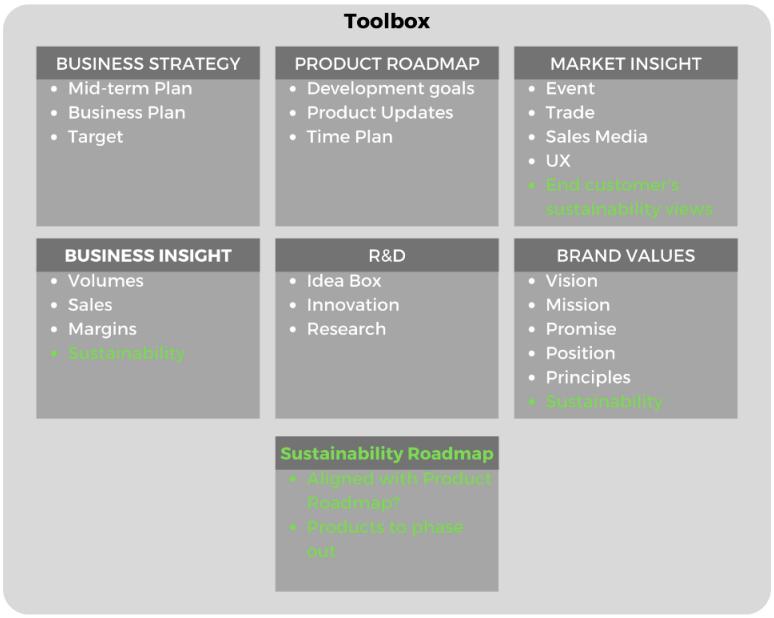


Figure 31. Version 1 of the *Toolbox* with added tools in green.

A new phase was added to this process named *Product Requirements*. Results from the information gathering phase showed that one of the most challenging steps of this process was setting requirements, especially sustainability requirements. Therefore, in this phase the Product Manager can specify measurable, specific sustainability requirements with the help of a Sustainability Manager or expert (see Figure 32). This phase is critical since it will dictate how sustainability will be considered during the development of the product (Luttropp and Lagerstedt, 2006). Finally, sustainability requirements and the updated *One-pager* were added in the output of this phase.

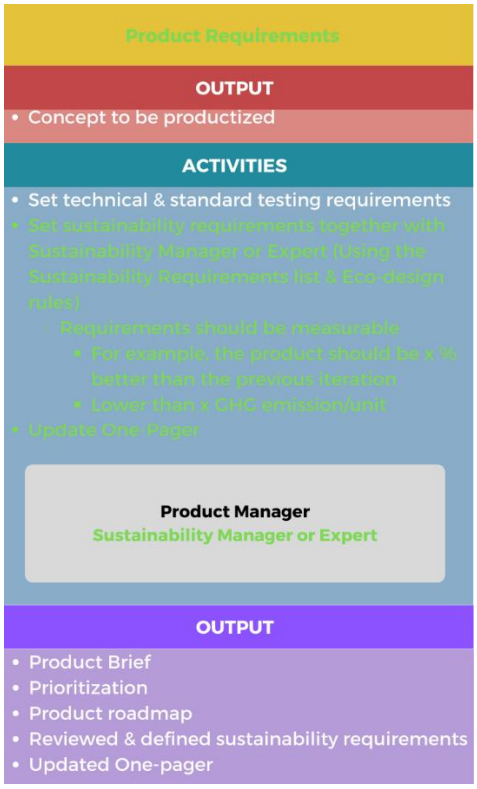


Figure 32. Added phase, *Product Requirements*, with points added in green.

3.4.1.4 Version 1. Development process

In Version 1 several steps and tasks were added into all the phases in the development process. See Appendix F for an overview of the changes made to the whole process.

Toll Gate 0

Before the DP begins, the product's requirements should have been set in the previous process (see Figure 33). Therefore, two discussion points were added. The first point is to check if the set sustainability requirements are measurable. The second point is to check if the *One-pager* has been updated.



Figure 33. Added discussion points to TG0.

Pre-study phase

In the *Pre-study* phase, a sustainability study was added (see Figure 34). The goal for the study is to make a rough estimation of the environmental impact of the product i.e., the GHG (greenhouse gases) emission per unit for the product and estimation of emissions from the testing that is to be performed.

Another step added to phase is to evaluate the set sustainability requirements and update them if it is deemed to be necessary with help of the findings of the sustainability study. Lastly, as Hallstedt et al. (2013) and Pujari (2006) points out it is important to have an environmental database and keep updating it for future development projects, any lessons learned from this phase should be documented. Eventually the experience and knowledge about sustainability gained over time will help Mips make even more accurate environmental claims and increase the working efficiency when dealing with complex sustainability problems.



Figure 34. Steps and tasks added to the *Pre-study* phase in green text.

Concept phase

In the *Concept* phase (see Figure 35) several concepts are to be created and in this phase the *Eco-design rules* were added as a design guideline to help the product developer design sustainable concepts. For example, one of the rules suggests the developer to use as few joining parts as possible, for a complete overview of the Eco-design rules see Table 1.

Moreover, the *Eco-design rules* were also added as a tool to evaluate the concepts. The *Eco-design rules* were chosen as an evaluation tool because it can be rewritten (Luttropp and Lagerstedt, 2006; Hallstedt et al., 2013) to work in a matrix or any other kind of evaluation method once Mips decide how they want to evaluate their concepts.

Another step added in this phase is the study and documentation of critical parts and their impact (Hallstedt et al. 2013) to pinpoint which parts have the highest GHG emissions. The idea here is that these parts should be locked by Mips, and the helmet brands will not be able to change the material and design specification that Mips have set since it can impact their overall GHG emissions.

Further, the authors added a step where the concept and supplier/manufacturer are evaluated from the sustainability point of view as suggested by Petersen et al. (2006).

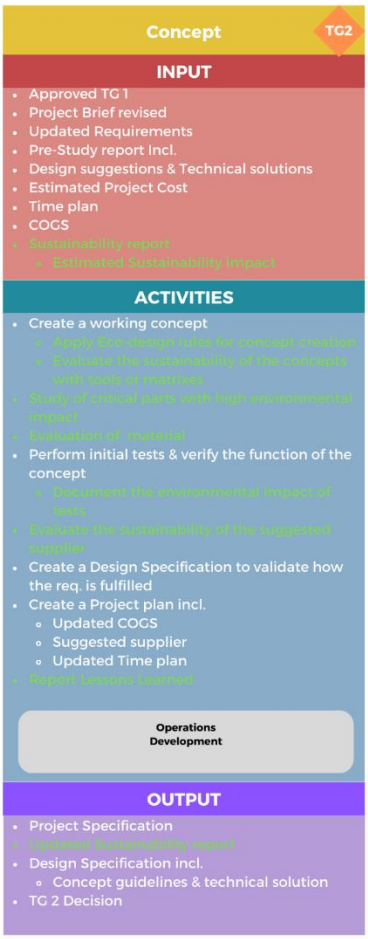


Figure 35. Steps and points added in the *Concept* phase in green.

Design phase

Steps added to the *Design* phase, shown in Figure 36, are documentation of the environmental impact of performed tests and verification of the sustainable requirements set for the product.

Another important step is the development of the marketing material that should include sustainable characteristics of the product as key selling point, and suggestions on how it could be presented in a desirable way to the helmet brands. As a salesperson it can be challenging to sell sustainable products without the relevant product facts being clearly stated and communicated by R&D. These key selling points should also be created in collaboration with a representative from the marketing and sales department.



Figure 36. Steps added to the *Design* phase in green text.

Pilot Production phase

Nothing was added in the *Pilot Production* phase since it was deemed that it would be too late to change anything at this phase. Moreover, all the product's issues should have been sorted out by this phase and if they had not been sorted out it would not have got past TG3.

Product Release & Conclusion

In the *Product Release & Conclusion* phase the creation of implementation guidelines for the product was added (see Figure 37). These guidelines should be created in collaboration with the implementation departments. The Implementation department has had more contact with the helmet brands and has more insights on the problems the customers might run into. By combining the knowledge of R&D and Implementation an optimal number of fixations could be set thus reducing material usage for the product.



Figure 37. Tasks added in the *Product Release & Conclusion* phase in green text.

Toll Gate 5

In TG5 discussion points, shown in Figure 38, for idea guidelines, recommendations for future project, and *One-pager* was added as natural effect process structure in TG5.



Figure 38. Discussion points added in TG5 in green.

3.4.2 Version 2

In this chapter the changes made to Version 1 will be presented as well as the development of the templates chosen by the process leaders during the feedback sessions. These templates are the *Sustainability Requirements list*, the *One-pager*, and the *Design log*.

One-pager

The second version of the *One-pager* had more detailed instructions on what kind of information should be written for each process (see Figure 39). The *One-pager* is used in all the processes as a handover document that contains the most relevant information. At the end of each process, the *One-pager* is updated with all the new information.

Innovation	Research & Pre-development	PM	Product development
Sustainability potential?	Sustainability research?	What sustainability requirements have been set? Which sustainability goals will the product contribute to?	Is the concept evaluated using the Eco-design Rules as a check-list?
Sustainability potential for idea	Result Sustainability report	Sustainability requirements	Evaluation results and Sustainability Leassons learned

Figure 39. Version 2 of the *One-pager*. First row in green are questions and second row answers or results.

During the IMP the sustainability potential of the candidate should be described. For the RPDP a first draft of what should be researched during this process and a summary of the findings should be written. In the PMP the sustainability requirements and goals should be defined and written. Finally, during the PDP the *Eco-Design rules* should be adapted to the product and used to evaluate the concept. The *Lessons learned* should also be summarized and written after the project is completed.

Design log

Basing the design on the file used by the process leader in the past, which had at this point been approved for general use in all processes, sustainability was added to this *Design log* as a column (see Figure 40). This will allow sustainability to be considered already when iterating the concept and finding a final working concept, instead of waiting until there already is a finished concept. The *Design log* will be stored in the *Summary file* as a sheet. The Design log is to be used during the concept development phases in both the RPDP and the DP.

Project name, #	Version	File name	Description	Image	Prod. Status, ETD	Tooling cost (USD)	Initial Part cost (USD)	Results, Comments	Sustainability results, Comments	Next Steps	Additional results/comments
Name	v1										
Further development ideas or thoughts after part completion:											

Figure 40. Sustainability results and comments added as a column.

Sustainability Requirements list

The *Sustainability Requirements list* presented is composed of the *Instructions* and the *Requirements table* shown in Appendix G. In the *Requirements table* the processes are presented in inverse chronological order. This means that the most current process will always be on top. Different types of requirements are required for the different processes. For example, for the IMP the requirements are to be vague. For each process that the project goes through the requirements become more specific and measurable. Therefore, for the PMP the requirements and the types of requirements in this process must be much more specific and are based on the key areas that the *Eco-design rules* covers (see Figure 41).

	Requirement 10	Product Manager	(n/a)		
	Requirement 9	Product Manager	(n/a)	Design for Dissassembly	User is able to dissamble product wth household tools
	Requirement 8	Product Manager	(n/a)	Design for Recyclablilty	Label for recycling // Use modular design for recyclable parts // Use breaking points to separate materials for recycling
	Requirement 7	Product Manager	(n/a)	Product maintenance	Machine washable product in 30 degrees with towels // Weather resistant
	Requirement 6	Product Manager	(n/a)	Product Life	Design for emotional durability // All parts in product shall have same life length
	Requirement 5	Product Manager	(n/a)	Material choice	0% non-recyclable material // Materials shall emit maximum X CO_2 t/ unit
	Requirement 4	Product Manager	(n/a)	Raw material	Use recycled materials // Use recyclable materials
	Requirement 3	Product Manager	(n/a)	Material usage	Padding usage X% lower than Product X
	Requirement 2	Product Manager	(n/a)	Resource consumption during production	Waste X% lower than product X // Produced in a factory with 100% renewable energy // Water usage per unit lower than product X //
Product Management	Requirement 1	Product Manager	(n/a)	Toxic substances	Toxic Substance usage X% lower than in product X

Figure 41. Types of Sustainability Requirements and examples in the white columns to the right.

In the *Instructions*, an explanation on how to write the requirements and what type of requirements are to be written for every process in chronological order is presented (see Figure 42).

Instructions

Innovation Management	**At this stage the only requirement is that the sustainability potential of the idea is described.** **The idea should contribute to at least one sustainabiilty goal that shall be identified.**
Research / Pre-Development	**In this stage the RPD developer will research sustainability aspects of the candidates given.** **Evaluate concept's sustainability by using the Eco-design Rules as a checklist.** **Make suggetions for sustainability requirements based on research findings.**
Product Management	**Evaluate the suggested sustainability requirements presented as output in the RPD process.** **Use the Eco-design Rules as a guide to update and define final requirements for the product.** **Incremental innovation products will have more specific requirements since they can be based on the LCA of the previous model.** **Radical innovation / New products will have requirements based on data estimations conducted in the RPD.**
Development Process	**The sustainability requirements are defined in the previous PM process.** **They are only to be updated if they need to be changed. The new requirements need to be approved in a gate.**

Figure 42. Instructions for the *Sustainability Requirements list*.

3.4.2.1 Version 2. Idea Management Process

For Version 2 only a template named *Idea box template* was created and added to the IMP. The purpose of this template was to have a standing template on the idea box that would explain to employees how to add an idea to the *Idea box*. The template and instructions on how to use it are shown in Figure 43. The template consists of a step-by-step guide on how to add an idea and text meant to be copied to a new card. Once copied in the new card, the questions should be answered by the idea generator if they already know the answers otherwise, they can leave some answers blank. The questions in the list ask users to shortly describe their idea, what purpose it has, what activity it belongs to, etc. The questions regarding the potential of the ideas should also be explained in a few short sentences and not rated as high, medium, or low. These questions were kept open to make the user shared as much information on the potential of the idea as possible which will later help to point the IM in the right direction. Further, since the idea generator and later the IM might not have all the answers at this stage, rating the potential (high, medium, low) would only be

an estimate with no backing behind it at this point. The more information the idea generator provides about their idea, the easier it will be for the IM to work with it.

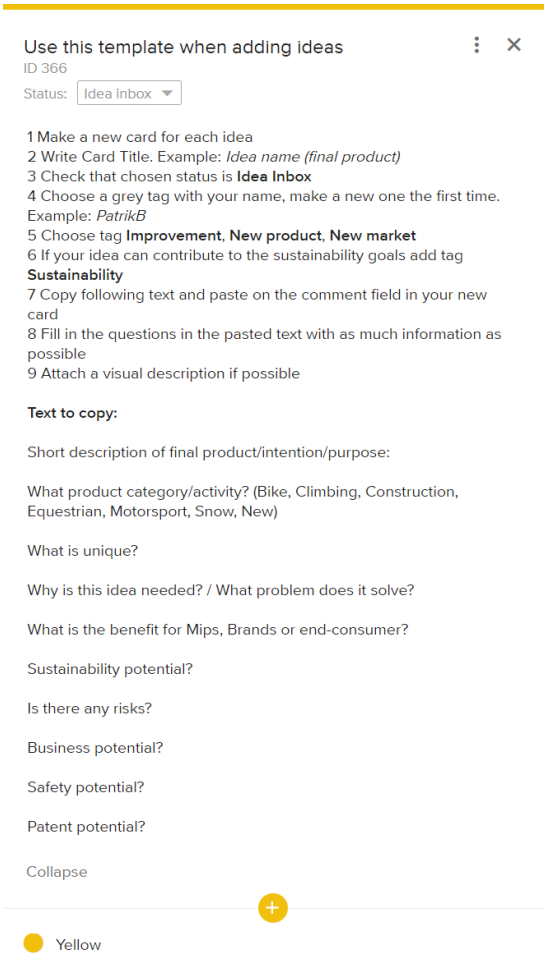


Figure 43. *Idea box template*.

Another task added to the Innovation Management process is for the Innovation Manager to use the *Sustainability Requirements list* to set vague sustainability requirements for the product for the selected *One-pagers* at the end of the second selection (see Figure 44).



Figure 44. Version 2 of the Second Selection phase.

3.4.2.2 Version 2. Research and Pre-Development process

Version 2 of the RPDP one differs from Version 1 in that the *Idea Brief* is clarified to have different characteristics when it comes directly from the IMP called *Candidate Brief* and when it comes from the PMP called *Product Brief* instead. The *Candidate brief* will have much less information and will therefore be vaguer while the *Product Brief* will be more specific (see Figure 45).



Figure 45. Version 2 of the *Idea handover* phase with clarification in cyan.

It was also noticed that the *Sustainability findings* would not be able to be stored in the *Design log* in the way that is needed since the *Design log* is individual to every project and therefore, not easily searched. The findings were in Version 1 presented as a part of the R&D report and suggested to be included in the *Design log*. According to research it would be more interesting to start building a searchable knowledge base with results of environmental impact and sustainability qualities of materials, designs, and components database (Pujari, 2006). Therefore, we suggest that an Excel file is started for materials and textiles, new materials and technologies, fabrication methods and other interesting sustainability findings can be stored on the relevant sheets. A database could be developed in the future with the information gathered with a better interface and a better search engine.

3.4.2.3 Version 2. Product Management process

Version 2 of the PMP was focused on the clarification of terms and ownership of the tasks in the process. In the *Input* brief the Idea management input was named "Idea management candidate" and it is also clarified that it is a part of the *One-pager* document. On the other hand, in the concept brief (Input from RPDP), it is clarified that the goals and sustainability requirements from the RPDP are a suggestion and part of the information provided in the *Summary File*. During the *Insights* phase the question whether the business model needs to change is brought up. Currently the owner of the business model definition for new markets and new products is not defined. Secondly, the recommendation of merging the Sustainability and Product Roadmap is made. This is because these two roadmaps need to be aligned and by merging them, both the Sustainability Group and Representative and the Product Manager will have an updated common tool to use (see Figure 46).

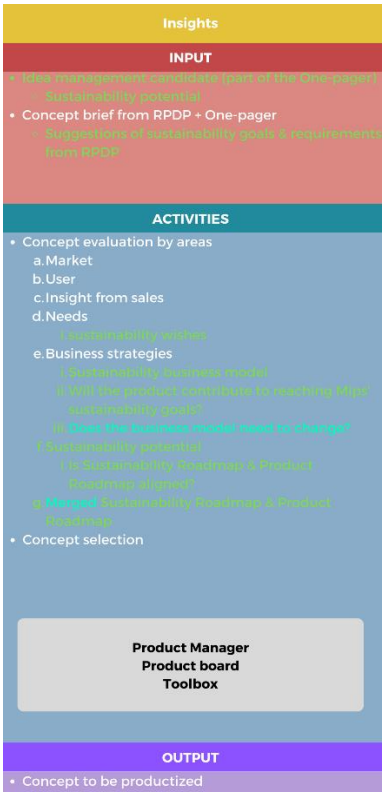


Figure 46. Version 2 of the *Insights* phase with changes in cyan.

In the *Product Requirements* phase it is clarified that the ownership of setting a clear sustainability prioritisation falls to management and that this should be done well before setting the requirements for the product, see Figure 47. During this phase, if a candidate that has come directly from the IMP does not contain enough information to be able to set concise and measurable sustainability requirements, the candidate should be sent to the RPDP so that *Sustainability Research* can be performed. In this phase the *Sustainable Requirements list* and the *Eco-design rules* are used.



Figure 47. Version 2 of the *Product Requirements* phase.

3.4.2.4 Version 2. Development process

In Version 2 the sustainability study that was supposed to be its own document in the Pre-study phase in Version 1, was made a part of the *Pre-study report* (see Figure 48). This decision to limit the number of documents was made to enhance the accessibility of knowledge and avoid scattering relevant information across multiple locations. Further changes made was to clarify that the results of the supplier search and evaluation is to be documented in the *Database template.*

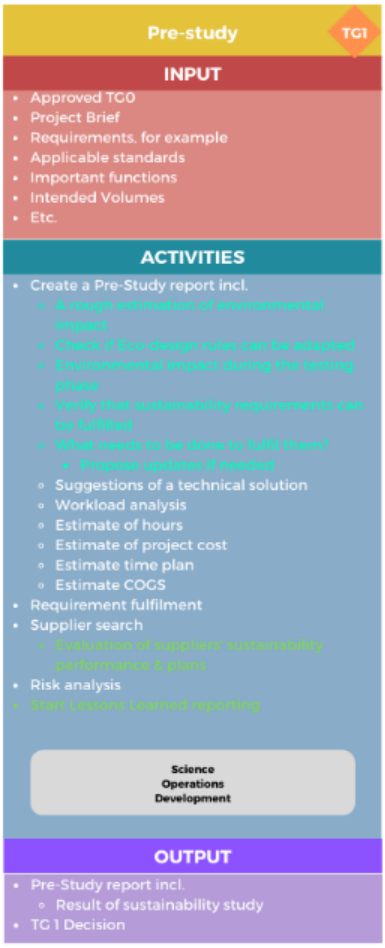


Figure 48. Version 2 of the *Pre-study* phase and changes in cyan.

Further changes made was to clarify the *Design log* is to be used to keep track of both positives and negatives sustainability implications for each iteration of the concepts. This change was done to the Concept and Design phase as show in Figure 49 and Figure 50.

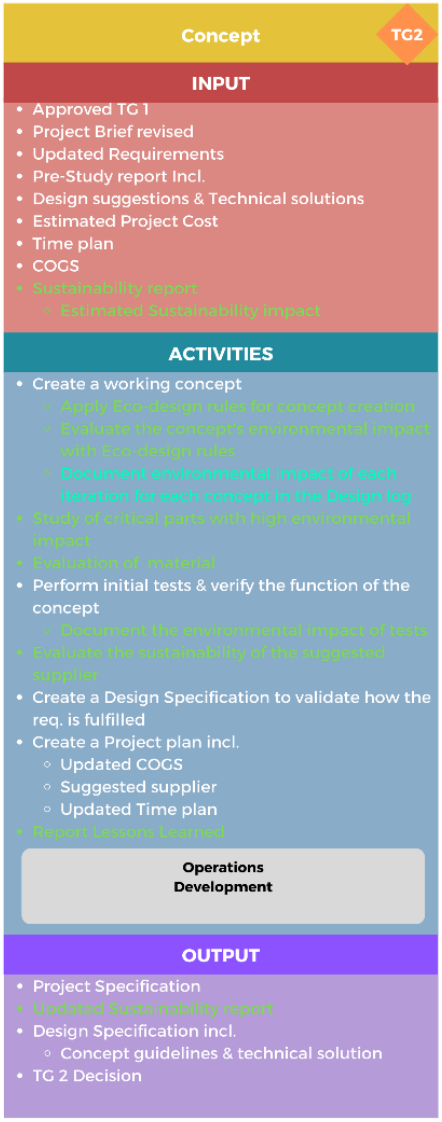


Figure 49. Second version of the Concept phase.



Figure 50. Version 2 of the Design phase.

3.5 Evaluation

3.5.1 Feedback session

The feedback of Version 2 was given at a meeting by every process developer and the head of R&D at Mips. In this meeting the solution for each subprocesses was presented and participants were asked to give feedback on the steps and activities added to the processes. Further, the templates created were also presented and here the participants were asked to comment on whether they understood how and where the templates are meant to be used and if anything was missing.

The feedback for activities added to the subprocess was to make the next iteration less clustered and clarify the intention steps or activities. Moreover, all participants agreed that the *Eco-design rules* were a great addition to their respective subprocess. However, the *Eco-design rules* could be applied in many ways depending on where the project is at and therefore it needed to be clarified visually how they were meant to be used.

The *Sustainable requirements list* template was received with positivity. One of the process leaders did however request it to be combined with the *Eco-design rules* to assist in the creation of requirements without having to change the window or search for them. The feedback for the *Idea box* template was to make it clearer for the idea generator that the questions in the template are optional and questions that cannot be answered can be left blank. Further, an *Unknown* category was requested for ideas that are hard to categorize in the *Idea box* template.

Further, one of the participants suggested that all the subprocesses should be combined into one larger process. Moreover, he pointed out that the usage of TG0 at different processes might confuse those who are not part of the R&D department.

3.5.2 Authors evaluation of feedback

After the feedback session, the added steps were revisited to ensure that a red thread existed throughout the whole process. Further, the templates created were also reviewed. As a result of these actions and the given feedback, it was realized that the solution needed to be improved to be made more cohesive. To achieve this, it was decided that the steps added need to be updated and simplified, if possible, for clarity. Further, since the *Eco-design rules* were a central part of the solution it was deemed that it should be clarified how they were supposed to be used at each step. Additionally, it was decided that the *Eco-design rules* and *Sustainability requirements list* should be merged based on the leaders' request and to compile the tools in the same place. Lastly, an *Unknown* category had to be added to the *Idea box* template.

3.6 Final Solution

In this chapter the new additions to Version 1 and 2 will be described. In Appendix H a complete summary of the Final Solution is presented in detail with all images of the processes.

The database

During the work in Version 2 a database where sustainability findings would be stored was proposed. A database will be of great value to designers and researchers alike and it would help with decision making during the Development processes (Pujari, 2006). To create a database, the first step would be to gather information regarding materials, suppliers, new materials and technologies, fabrication methods, and important sustainability findings.

In line with Mips' current work structure, an Excel file could be developed in the meantime until a formal database with a better search engine and UX-design can be set in place. Some of the material specialists already have their own versions of such a file, where they store the latest information regarding the materials they have studied.

For this final version of the framework the authors propose a suggestion on how this Excel file could look like. The information in this file would be added by the material specialist and operations specialist only, this to make sure that correct and updated information is always available. Each sheet could be password protected so that only the specialist in charge can edit them while everyone in the organisation / R&D department can consult it.

The suggested template should be stored in an easily accessible folder in the shared R&D documents. As a starting point four sheets are suggested: *Materials*, *Fabric-Foam*, *Suppliers* and *New Technology*. The different sheets' columns should be tailored to the specific needs of each topic. For example, the *Fabric-Foam* sheet was taken directly from the Textile Specialist's own file an includes very textile-specific data (see Figure 51). The *Materials* sheet was based a draft provided by the Material Specialist. The *Supplier* sheet information was based on input gathered during the *Empathise* phase. In *New Technology*, general information regarding new technology, sustainability findings and other interesting information should be stored. The complete suggested template can be found in Appendix I. To find the exact information that should be in the future database a future analysis should be conducted together with the material and operations specialists. This cannot be done within the scope of the project and therefore, this template is only a suggestion for Mips that can act as a starting point would they choose to develop the database.

Mi (Mips), K (Knitted), W (Woven), N (Non woven), F (Foam), I (insulation) W (Woven)							*Stretch properties:*		Link to test					Solution dye / recycled / oeko-tex dye / Sorons fiber					
Type	**Art. Number**	**Fabric name**	**Replace**	**Weight / Thickness**	**Fabric Width**	**Col.ways**	**Artical no. Supplier**	**Supplier**	**Composition**	**OLD replace**	**Finish**	**Test reports / Data sheet**	**Washing instruction**	**Properties Marking**	**Sustainable Marking**	**MOQ**	**Price(RMB/ Yard)**	**Notes**	**Marking**
Knitted	XX X XX	Name		140g/m2	62"	Black Dark grey	XXXXXX	XX	*88% RECYCLED POLYESTER 12% ELASTANE*	*90% Polyester* *10% Spandex (Elastane)*	N/a	Link	Machine wash 30°C Do not bleach Mild drying processes Iron at low temperature Do not dry-clean	Moisture-Wicking Recycle polyester	Oeko-Tex Blue sign	Min Qty / order 3000 yard. Min Qty/ color 1000 yards	3,05US/Yards		

Figure 51. *Fabric-Foam* sheet in the *Database* template.

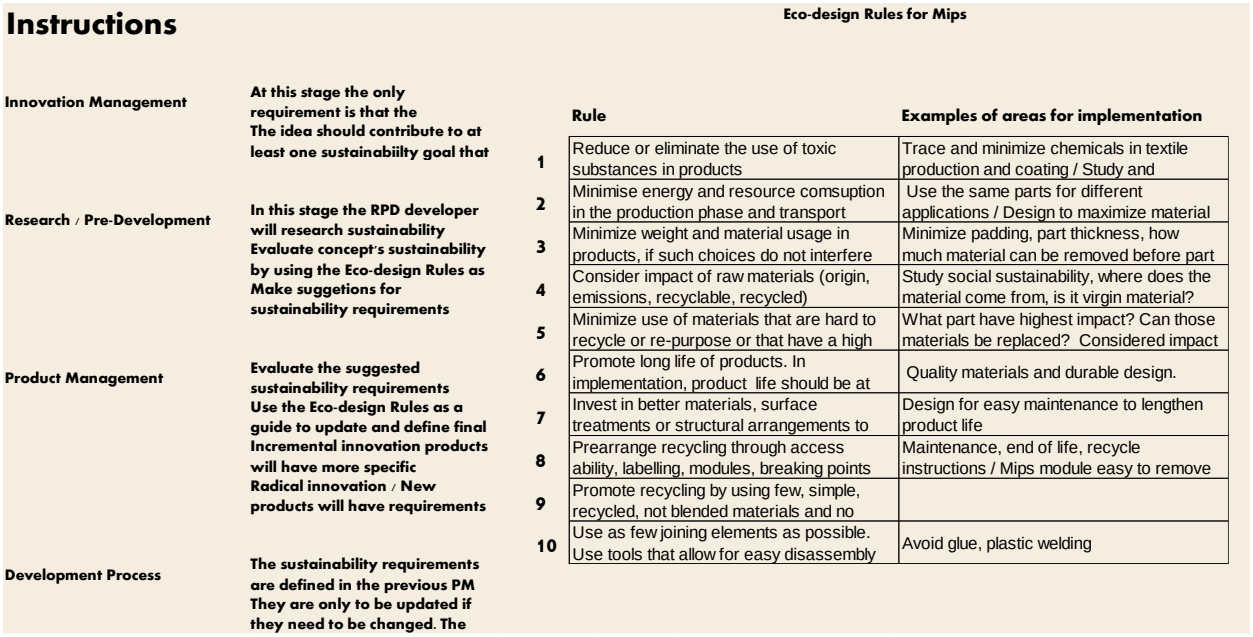
The Database excel file template is to be used as a working document for the Material Specialists and Operations specialists. Already in the IMP the document can be consulted when defining the sustainability potential for the idea candidates. In the RPDP this document can be both consulted during the process and later updated with the sustainability findings at the end of the phase. During the PMP the document could be consulted while defining the requirements. Finally, during the DP this document can be consulted when making decision regarding the concept, materials and production techniques and later updated after the project's conclusion with any learnings on the materials or sustainability aspects.

Requirements list + Eco-design rules

Feedback from the Version 2 presentation questioned where the *Eco-design rules* should be stored for easier access. A document with the *Eco-design rules* should be stored in an easily accessible folder in the R&D common documents folder.

The *Sustainable Requirements list* is the only template that requires the use of the *Eco-design rules* when developing and writing down the *Sustainable Requirements*. Therefore, the *Eco-design Rules* were placed to the side of the *Instructions* section so that they could be consulted when writing the *Sustainable Requirements* during the RPDP and PMP (see Figure 52). This template is

used during all the phases since the requirements are updated at the end of the IM, RPDP and PMP as well as during the DP if updated to the product requirements are needed.



Instructions

Innovation Management

At this stage the only requirement is that the The idea should contribute to at least one sustainabiilty goal that

Research / Pre-Development

In this stage the RPD developer will research sustainability Evaluate concept's sustainability by using the Eco-design Rules as Make suggetions for sustainability requirements

Product Management

Evaluate the suggested sustainability requirements Use the Eco-design Rules as a guide to update and define final Incremental innovation products will have more specific Radical innovation / New products will have requirements

Development Process

The sustainability requirements are defined in the previous PM They are only to be updated if they need to be changed. The

Eco-design Rules for Mips

	Rule	Examples of areas for implementation
1	Reduce or eliminate the use of toxic substances in products	Trace and minimize chemicals in textile production and coating / Study and
2	Minimise energy and resource comsuption in the production phase and transport	Use the same parts for different applications / Design to maximize material
3	Minimize weight and material usage in products, if such choices do not interfere	Minimize padding, part thickness, how much material can be removed before part
4	Consider impact of raw materials (origin, emissions, recyclable, recycled)	Study social sustainability, where does the material come from, is it virgin material?
5	Minimize use of materials that are hard to recycle or re-purpose or that have a high	What part have highest impact? Can those materials be replaced? Considered impact
6	Promote long life of products. In implementation, product life should be at	Quality materials and durable design.
7	Invest in better materials, surface treatments or structural arrangements to	Design for easy maintenance to lengthen product life
8	Prearrange recycling through access ability, labelling, modules, breaking points	Maintenance, end of life, recycle instructions / Mips module easy to remove
9	Promote recycling by using few, simple, recycled, not blended materials and no	
10	Use as few joining elements as possible. Use tools that allow for easy disassembly	Avoid glue, plastic welding

Figure 52. N*ew Sustainable Requirements list* instructions with *Eco-design rules.*

Checklist template

Mips requested that the steps regarding this project (sustainability steps) were added to their *R&D Process Checklist* sheet in their *Summary File*. The *R&D Process Checklist* is a document where the different steps that will be taken in each process are listed and is meant to be used as a checklist by the Project leader during the project. After our work, it currently also includes the sustainable steps of our framework. Since this is an internal working document, it will not be shared in this report.

4 DISCUSSION AND CONCLUSIONS

In this chapter, the final discussion and conclusions will be presented.

4.1 Discussion

Purpose
The focus of this thesis was to map current practices and develop a framework that would add sustainability to their new Development Processes. The authors together with Mips, deem that the purpose of this thesis was achieved.

A practical thesis
The final framework was, tailored to Mips' needs, current processes, and organisational structure. To achieve this, the development of the framework was conducted as a product development project. Therefore, as commented in Chapter 2, the *Design Thinking* methodology used was a perfect match since it catered to needs and requirements of the R&D department at Mips. The user of the product (framework) was studied, and their current practices mapped. Once the user (Mips' R&D department) and their practices were understood, a *Needs and requirements* list was developed.

Since Mips was developing their new Development Processes framework at the same time this thesis was conducted, and because our work regarding the sustainability aspects of their new framework was to be implemented immediately, this thesis had since the beginning a very practical nature. Some of our ideas, such as having a sustainability representative in the *Decision team* were already put in motion after the first concept presentations. Additionally, the authors could receive continuous feedback from the interested parts at Mips on the challenges that arose when implementing our recommendations. This allowed to make changes while the development process was still underway. For example, the *Idea box template* was delivered to Mips by updating it directly into their *Idea box*, where it would be put to use right away.

On the other hand, the fact that Mips was parallelly still developing the Development Processes meant that some information, and specially the working documents, were in continuous development. Some documents used to base our templates on had already been updated by the end of this thesis. For example, the *One-pager* was redesigned based on changes on how it was going to be used by Mips. This re-design did include some of the aspects that were suggested by the authors, such as having sustainability included as more than a grade or potential. Sustainability was instead included as a main part of the template where more information would be updated and filled out as the project progressed through the processes.

The final solution
The final solution is a framework largely based on Mips' needs as a company, but also the theory presented in the *Literature Study* results chapter. In Chapter 3.1, the relevant literature that was the base of our brainstorming and later concept generation, was presented. Therefore, our solution reflects the theory quite well. The authors have taken the aspects of theory regarding the process,

people and tools that better met the needs and requirements and that could therefore be applied to Mips.

Firstly, the sustainability aspects are integrated and considered from the first step of the Development Processes based on recommendations by Bovea and Pérez-Belis (2012), Luttropp and Lagerstedt (2006), Ny et al (2008) and Sherwin and Bhamra (1999). In order to steer the product development process from the start towards a sustainable product as Luttropp and Lagerstedt (2006) propose, the sustainability requirements of the product were introduced as its own step and as a *Sustainable Requirements* list. Secondly, the involvement of a sustainability manager or coordinator in the product development process as proposed by Ritzén (2000) was also implemented throughout the processes and specially in the Toll gates by adding a sustainability representative to the Decision team. The knowledge that this person should hold was partly based on Mips' needs for this role and on recommendations by for example Hallstedt et al. (2013). Further, the earlier supplier involvement and evaluation added to the beginning of the Development Process that would help the informed decision making regarding the impact of manufacturing was based on input from Pujari (2006) and Petersen et al. (2005). The transparency required for these relationships to work as stated by Holtström et al. (2019), will also be crucial for the further development of sustainability work at Mips. Finally, one of the latest additions to the concept was the recommendation to start gathering information for a future database that would support the product developers in achieving more sustainable products as recommended by Pujari (2006), Hallstedt et al. (2013) and Ritzén (2000).

Secondly, Mips, being a growing and innovative company, has specific needs regarding flexibility and autonomy in their R&D department that had to be respected and conserved to the maximum extent possible for them to be able to continue innovating as they have. This flexibility requirement is reflected in our framework in several ways. The sustainability framework needed to be simple to understand and the tools easy to successfully implement as stated by Ritzen (2000). The methods chosen needed to be adaptable to the different types of projects that it would be implemented on. This meant that for example during the concept development and evaluation, simple and adaptable methods needed to be used to evaluate sustainability. Most of the methods found in literature were based on LCA calculations that required a lot of information were time-consuming and were complicated to implement (Bovea and Pérez-Belis, 2012). It was deemed that such data-based methods were not the best solution for Mips. Since Mips is in the beginning of their sustainability work, not enough information and resources were available for such a data-heavy quantitative methods to be used for all the products and projects. Moreover, Mips does not produce any of their products themselves, or owns any of the manufacturing facilities where their products are made. This means that they can't control most aspects of the production and are not in possession of all the information at early stages of the concept development process when suppliers are not yet chosen. The Operations department and the material specialists are still currently analysing their suppliers and trying to find suppliers transparent enough that can provide all the data necessary to calculate the impact of a product. In the future, once these suppliers are identified, they could support Mips with data during the development process and be involved earlier on so that informed decisions can be made already in the concept phase. Therefore, and to make sure that the proposed method was general enough to fit the framework and be applied to all types of projects and products, the *Eco-Design rules* were chosen as the proposed general method to evaluate sustainability at different stages in the processes. These rules were customized to fit Mips, as recommended by Luttropp and Lagerstedt (2006).

Further, during the *Ideate* phase it became clear that the involvement of team China would need to be set out of scope. Even though the issue of their involvement was a part of the *Needs and Requirements* list, it turned out to be a much more complicated topic and it was apparent that the authors lacked information regarding Mips operations in China. Time constraints meant that it was not possible to iterate the *Empathize* phase and gather further information necessary to provide a clear understanding of the issue and develop a solution. Moreover, the process leaders' framework barely names team China or provided more information about their involvement. Therefore, together with the Industrial Supervisor, the topic was deemed to have a lower priority than the rest of Mips' needs that directly related to the Development Processes.

4.2 Conclusion

The sustainability framework developed and presented in this report does achieve the purpose of this thesis in providing the steps and tools necessary for Mips to include sustainability in their new Development Processes at the R&D department. Their current practices were successfully mapped, and a tailored framework developed, and Mips' needs and requirements defined and later used to develop the final solution presented in Appendix H. Since the Development Processes were still in development, the final templates needed to be based on the information gathered before and during the *Ideate* phase. This means that changes made after this point by the Mips' process leaders could affect the relevancy of some aspects of the Final solution.

5 RECOMMENDATIONS AND FUTURE WORK

In this chapter, recommendations on more detailed solutions and future work in this field will be presented.

5.1 Recommendations for the case company

The first recommendation, for Mips, is to create a database in the near future. To create this database the material, textile, and operations expert should be consulted to gain an insight on what information they think should be stored in the database. The goal for this database is to store and organize relevant knowledge and best practices within Mips. Not only would a database or a knowledge management system improve sustainability practices, but it would also provide other benefits such as:

1. **Improved Sustainability Practices:** A knowledge database can help sustainability practices by gathering and distributing information about environmentally friendly manufacturing processes, design principles (such as *Eco-design rules*), and materials. It also supports product developers by giving them access to insights regarding sustainable practices, guidelines, and regulations. This in turn mean that sustainability is being taken into consideration during the design and development of the product.
2. **Knowledge Sharing and Collaboration:** A database facilitates sharing knowledge across different teams of the company that is involved in the product development. This allows employees to access and share a collective pool of knowledge that encourages innovation and enables for more efficient decision making during the whole development process.
3. **Learning from Past Experiences:** Documenting previous development experiences, success, failures, and lessons learned in a database can help Mips to avoid repeating past mistakes. The designers, developers, and other stakeholders would be able to use past data to identify patterns, customer preferences, and make informed decisions.
4. **Efficient Problem solving:** Since a database can provide access to guides, technical specifications, and solutions to past issues, it can be used as resource when faced with for example, difficult sustainability problems. Furthermore, this reduces the time of troubleshooting thus leading to an overall efficient product development process.
5. **Continuous Improvement:** By documenting product performance, customer feedback, and market trends, the database can help Mips to identify opportunities for improvements. In the long run it can help Mips stay competitive and respond accordingly when customer needs and the market evolve.

The second recommendation is for the R&D department to set short-term goals or milestones in the same way that Mips as an organisation has a GHG reduction target for each year. These goals are meant to help the R&D department to navigate and know what they should prioritize in the upcoming 1, 3, and 5-years. Since the organisations goals are extremely ambitious and numerous members of the R&D department have expressed that it is unclear what they should personally and as a team to reach these goals. The milestones can than help these employees to start working

towards the overarching goals without by focusing on their department's smaller goals and milestones.

The third recommendation is for the R&D team to run a test of the suggested framework on a real project at Mips, use the templates and makes changes that is deemed necessary or more practical.

The last recommendation for Mips would be to further study Team China's involvement in the Development Processes. Since they are in contact with suppliers and are a part of testing the prototypes, they could have an impact on Mips' sustainability impact and therefore, be a valuable resource to improve sustainability in products and processes.

5.2 Recommendations for companies

For companies that are in the beginning of their sustainability journey the authors would recommend:

To consider environmental sustainability at the early stages of the product development process since it can be costly and time consuming to add sustainability requirements in the later stages of the process. Further, they should set one big sustainability goal, as Mips has done, for the whole organisation and break it down into smaller achievable goals. Lastly, they should start measuring and document their sustainability impact. Companies that start measuring their sustainability impact will overtime gain an understanding of how their operations affect the environment and will gain an insight into where improvements should be made. From there, they should start to make small changes in order to become more sustainable.

5.3 Future work (Framework)

Since a development project at Mips can take between 6 to 36 months and the thesis's time limitation is 20 working weeks, the final solution could not be tested in project. Therefore, the future work of this thesis is to first and foremost to be tested on a pilot project at Mips. It could also be tested at a company of similar size and ambitions at the beginning of its sustainability journey if their needs match Mips' so that the framework can be applied. The aim of this test should be to find out what works in practice and update the framework accordingly.

Moreover, because the framework has centered around how Mips works and how their new product development would work in practice, a potential future work would be to generalize the framework to be applied to other companies. Currently, the framework is tailored to work in a phase-gate product development model. It might be intriguing to study if it could be applied to other product development processes such as agile product development.

Furthermore, since Mips is a growing company at the beginning of their sustainability journey, they are lacking in some knowledge regarding sustainability practices. It could also be of interest to customize the framework for well-established firms and small companies that completely lack sustainability practices and goals.

Since this thesis has had environmental sustainability of the product development process as its only focus, the framework does not cover the social and economic aspects of sustainability. It could therefore be of interest to consider these aspects to expand the framework and give it a holistic perspective of sustainability.

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APPENDIX A: INTERVIEW GUIDE

The interview guide used in this thesis is presented below and is divided into different topics. Every question was not asked to all the interviewees, only relevant ones for the respective role.

General questions

- Could you state your name and tell us a bit about your background?
- Just for the record, could you please tell us your job title and how long have you been at Mips?

Job Questions/Old process

- What are your responsibilities?
- What does your day-to-day look like?
- Could you reflect upon one or two things you love about your job?
- And now, on the other hand, is there anything that would like to improve or change that would make you more satisfied with your role?
- What is your position or responsibility in a project? (Project manager, etc.)
- Can you choose the projects you work on or are your projects assigned to you?
    - How much freedom do you have to make your own decisions during projects.
    - When you make decisions are they based on your own judgement, or do you follow a set guide?
- You are currently defining a new Product Development Process, but how have you/or your team (personally) worked until now?
    - Have you used the same process or has your way of working changed from project to project?
    - Do you have your own process?
        - Could you explain the steps?
        - Do you have specific phases?
    - In your old process, has sustainability been considered?
        - How so? Did you manage, measure, or prioritize sustainability?

Sustainability Questions

- What is your personal view on sustainability?
- Are you familiar with Mips' sustainability goals and strategies?
    - Do you think your personal view aligns with Mips' sustainability goals?
    - Do you think Mips' sustainability goals are enough? Do you see any potential improvement?
    - Do you think Mips is calculating the correct things? Is there any improvement that could be made to this?
    - Do you think all your co-workers know where they can find Mips' sustainability goals and policies?
    - What business opportunities do you think the sustainability goals can provide for Mips?
    - Do you think there is enough knowledge within Mips to achieve the sustainability goals?
    - Can you think about anything specific that needs to change to achieve them?

    - Since signing the SBTi goals, have you seen a change in the way of working?
- How do you choose materials? Do you have any tools? What are the material decisions based on?
- Do you feel that your decisions have an impact on Mips environmental sustainability?
    - How so?
    - Are there any frustrations or challenges that you regularly face in your role related to sustainability?
- If it were up to you, what would you have the R&D do to achieve Mips sustainability goals.
    - How could you contribute to this?
- What would you or co-workers in your role need to improve your sustainability work?
    - Any tools, resources, directions, guidelines, knowledge needed?
    -

Materials

- Who chooses materials? Who is involved in this decision?
    - How is the decision taken?
    - When in the process do material recommendations happen?
    - Can the helmet brands set requirements on materials chosen on the Mips' products?
- What aspects are taken into consideration when choosing a material? Just the mechanical qualities or other requirements like possibility to disassembly, or requirements for assembly?
- Do you know what environmental impact a material choice will have? Is there a database? Do you have a list of materials that are pre-approved from a sustainability point of view? If not, would you like to work like that?
- Is there a database where the materials and impact are stored? Do you store material knowledge from previous projects?
- Who's responsible for knowing how the material is produced? Material specialist or operations?
- Do you take into consideration the amount of material that will be needed in the final product when choosing a material?
- Do you know what factory that component will be produced in? And if it uses green energy?
- Scope 3 as stated in the SBTi relates to manufacturing and external operations that cannot directly be affected by Mips. How could your department affect and improve the impact of these operations?

Science Based Targets (SBTi)

- Could you briefly describe the structure of the SBTi-group and its sub-group?
    - How is information from this group spread to the different departments?
    - Who is this group responding to?
    - Is everyone in the group on the same page? Goal wise?
- Could please explain the SBTi goals, scopes 1-3, with your own words?
    - Do you feel that the Sustainability group is doing enough?
        - Could you elaborate? Why or why not?

APPENDIX B: THE RESEARCH AND PRE-DEVELOPMENT PROCESS

Overview of the subprocess *Research and Pre-development process.*

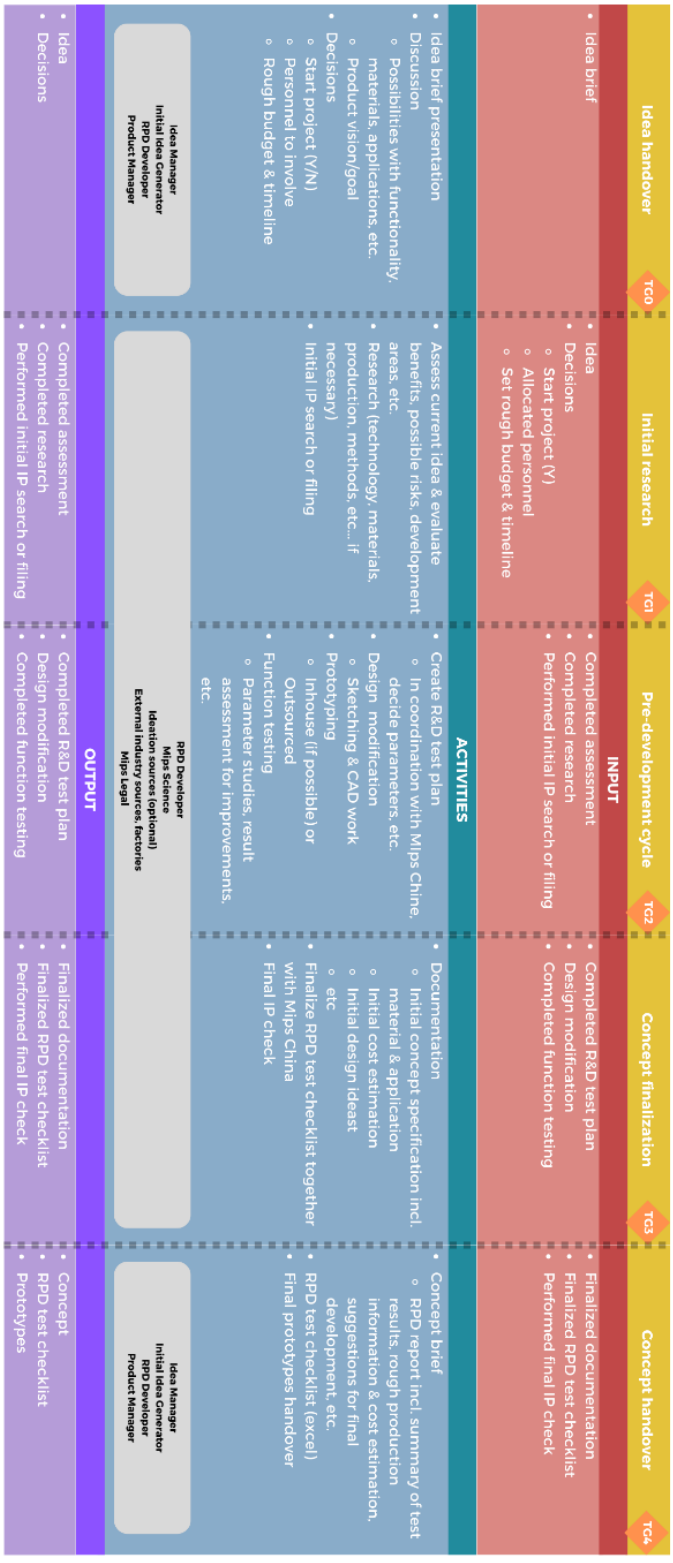


Figure. Overview of the current *Research & Pre-development process*

APPENDIX C: THE DEVELOPMENT PROCESS

An overview of the development process that is under construction.

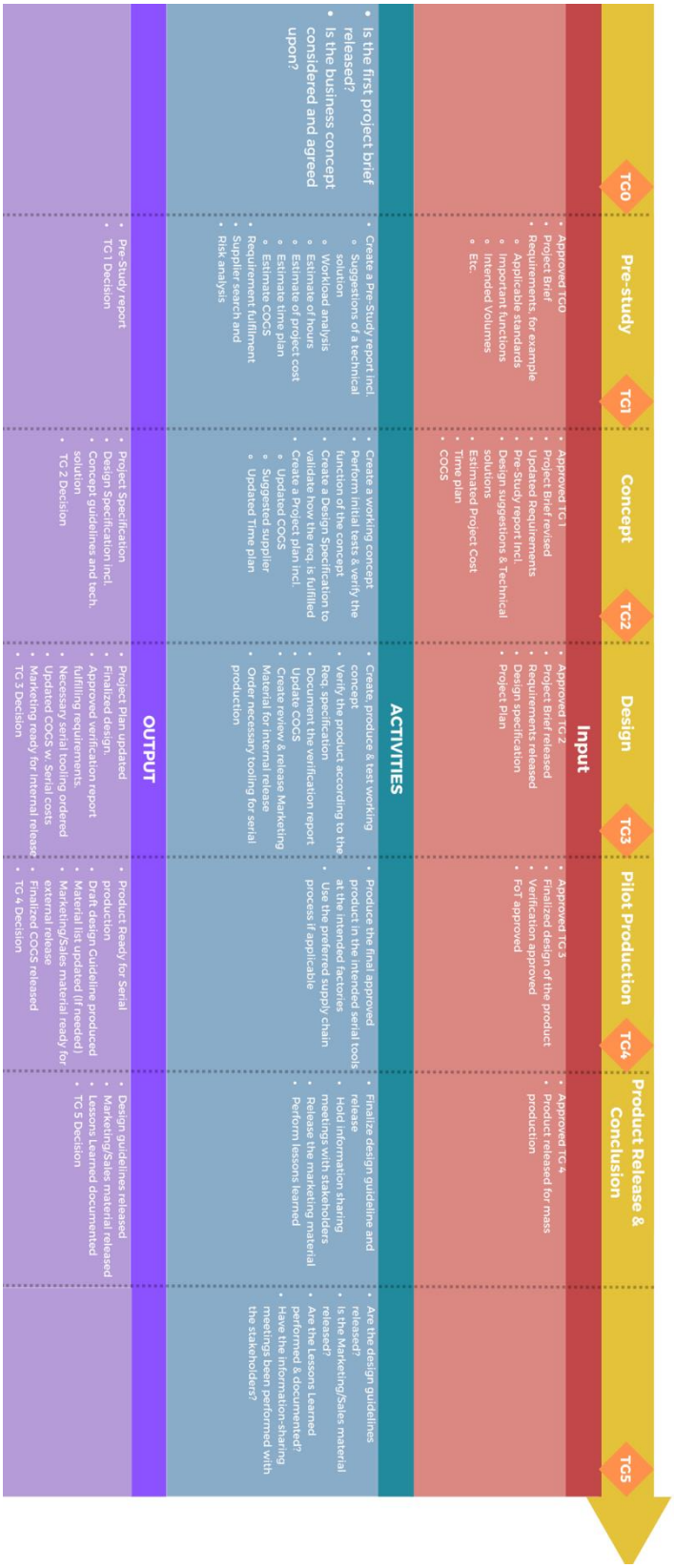


Figure 1. Overview of the *Development process*.

APPENDIX D: NEEDS AND REQUIREMENTS LIST

This chapter presents a table of the needs and requirements list.

Input (Interview / observation)	Need	Final requirement	Prio 1 or 2 or 3, Suggestion, Out of sco
Not everyone thinks about sustainability in their daily work	Ingrain sustainability on the company's culture	Remind continuously of sustainability during processes	1
No specific guideline on how to handle sustainability in projects	Support for sustainability decisions and actions	Provide clear guidelines on how to work with sustainability during projects	1
Lack of environmental thinking in PDP	Promote environmental thinkinng in PDP	Define what sustainability requirements are needed at each stage of the processes	1
Too late if the sustainability requirements are set when development of the product has started	Sustainability requirements should accompany the project from idea through all the processes	Provide guidelines for defining sustainability requirements already at the begining of the Innovation Management process	1
Current evaluation of concepts has no set method, some decisions are taken in collaboration with project leader or head of product development but many decisions are up to the developer.	Evaluation of concepts before moving forward (sustainability + other factors)	Concept's sustainability shall be evaluated	1
Interview with PM	Stakeholders involved in product management process require support in understanding sustainability goals to make the right decisions on products to develop / Stakeholders involved in product management process require specific sustainability data that can be added in the product brief	Provide additional support to product management regarding sustainability (goals, specific data)	1
It is too late to set requirements during the Development Process, the project brief should already include measurable requirements	Promote setting product requirement before DP	Sustainability requirement shall be measurable before the Development Process	1
Innovation might be hindered if too strict requirements are set in the early processes	Have vague requirements in the first subprocesses	Sustainability requirements for non-incremental projects shall not hinder innovation during Innovation Management and Pre-development processes	1
In each process the knowledge about sustainability aspects increases, this means that sustainability requirements can be refined after every process	Document and sustainability define continuously	Sustainability requirements shall be refined during every process	1
In order for sustainability to be present throught all the processes, some requirements need to accompany the project already from the begininng	Document and sustainability define continuously	Sustainability requirements shall accompany the project from idea to conclusion	1
No set practice on checking after-life impact in current PD process	Increase Mips' knowledge of after-life impact of their products	Consider after-life impact of products early during the product development processes	2
In key parts with high environmetal impact, materials should not be restricted and this comunicated to the customer so that Mips has control over the sustainability impact.	Increase control over the final material choice of the product - R&D	Restrict customer changes of final material choices	2
Team China informed late on progress of early project stages	Involve the China team earlier in the PD process	Involve at least one quality department member in China from the begining in the pre-development and development processes	2
Some customers have been known to want to change materials after the product has been developed, this could lead to negative environmental impact since Mips r&d team's selection should have been based on sustainability aspects. Therefore the parts with high environmental impact due to material or size should be identified during the development process.	Document material impact	Identify key parts with high environmental impact	2
The team in China is out of the loop	Improve feedback loops with China team during the whole PD process	Team China shall be included in important briefings throughout the Pre-development and Development processes	3
R&D currently working without information structure	Need for a standard way of organising information	The solution should give easy access to information during and after projects	3
Hard to know what the suppliers are capable of	Easy access to supplier capabilities and sustainability performance	The solution should be able to track supplier capabilities and performance during and after projects	3
Interview with R&D	Improve the communication between departments	The solution should allow users to share information between different departments	3

Figure 1. Screenshot of the needs and requirements list.

APPENDIX E: VERSION 1 OF THE RESEARCH AND PRE-DEVELOPMENT PROCESS

The figure below shows an overview of changes made to the *Research and Pre-development process.*

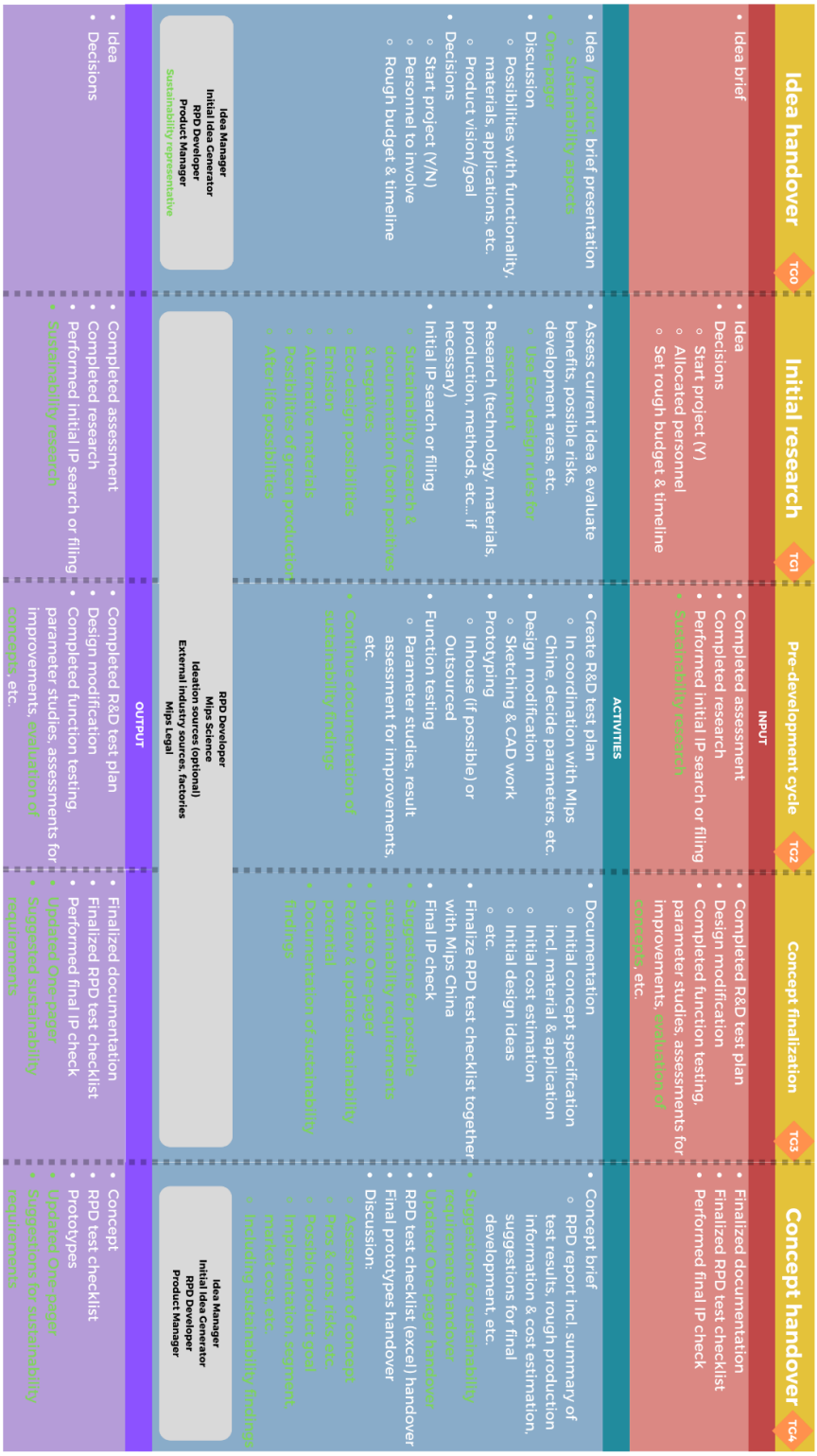
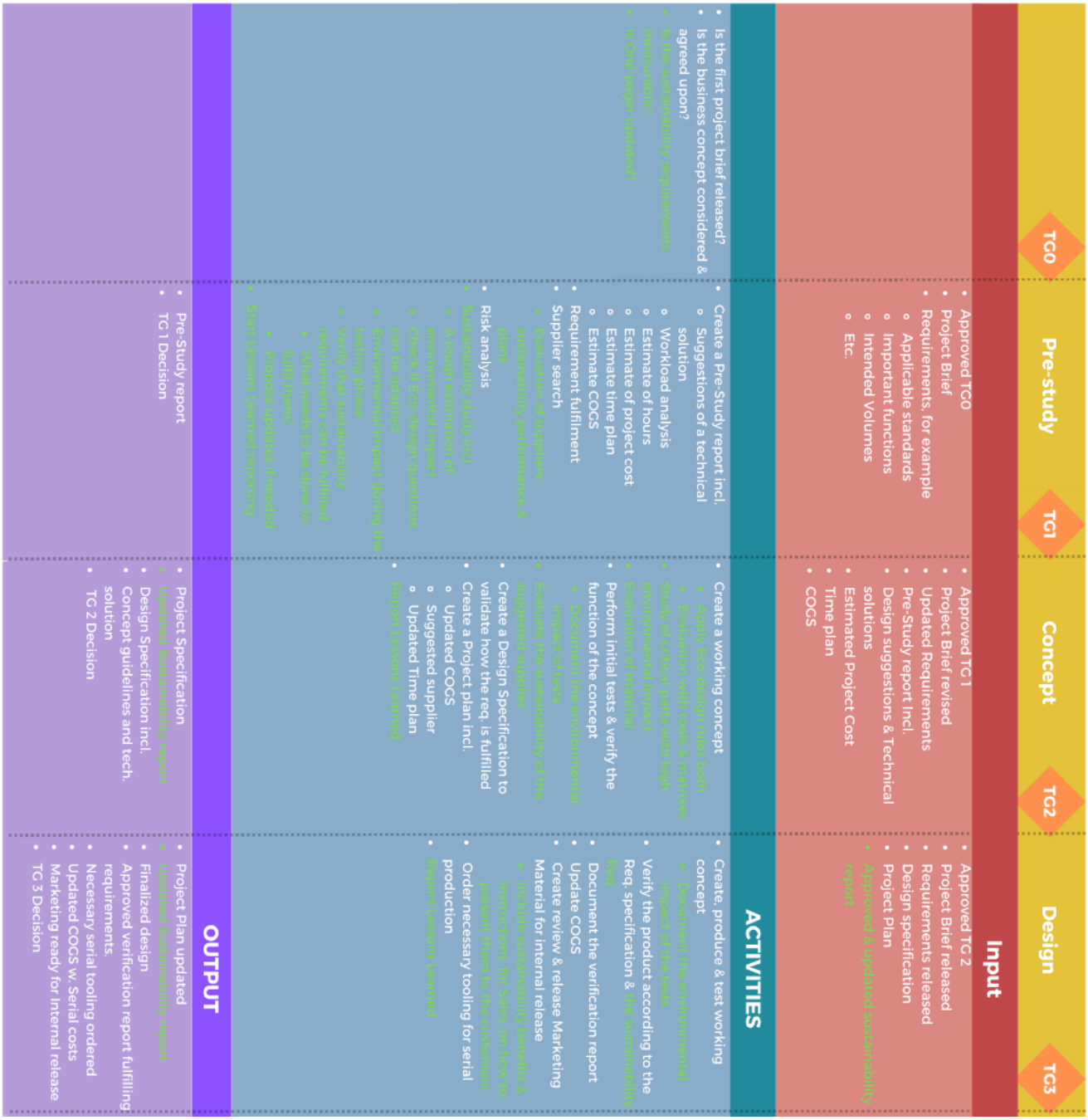


Figure. 1 Overview of version 1 for the *Research & Pre-development process*

APPENDIX F: VERSION 1 OF THE DEVELOPMENT PROCESS

An overarching view of changes made to the *Development Process.* Changes made are shown in Figure 1.



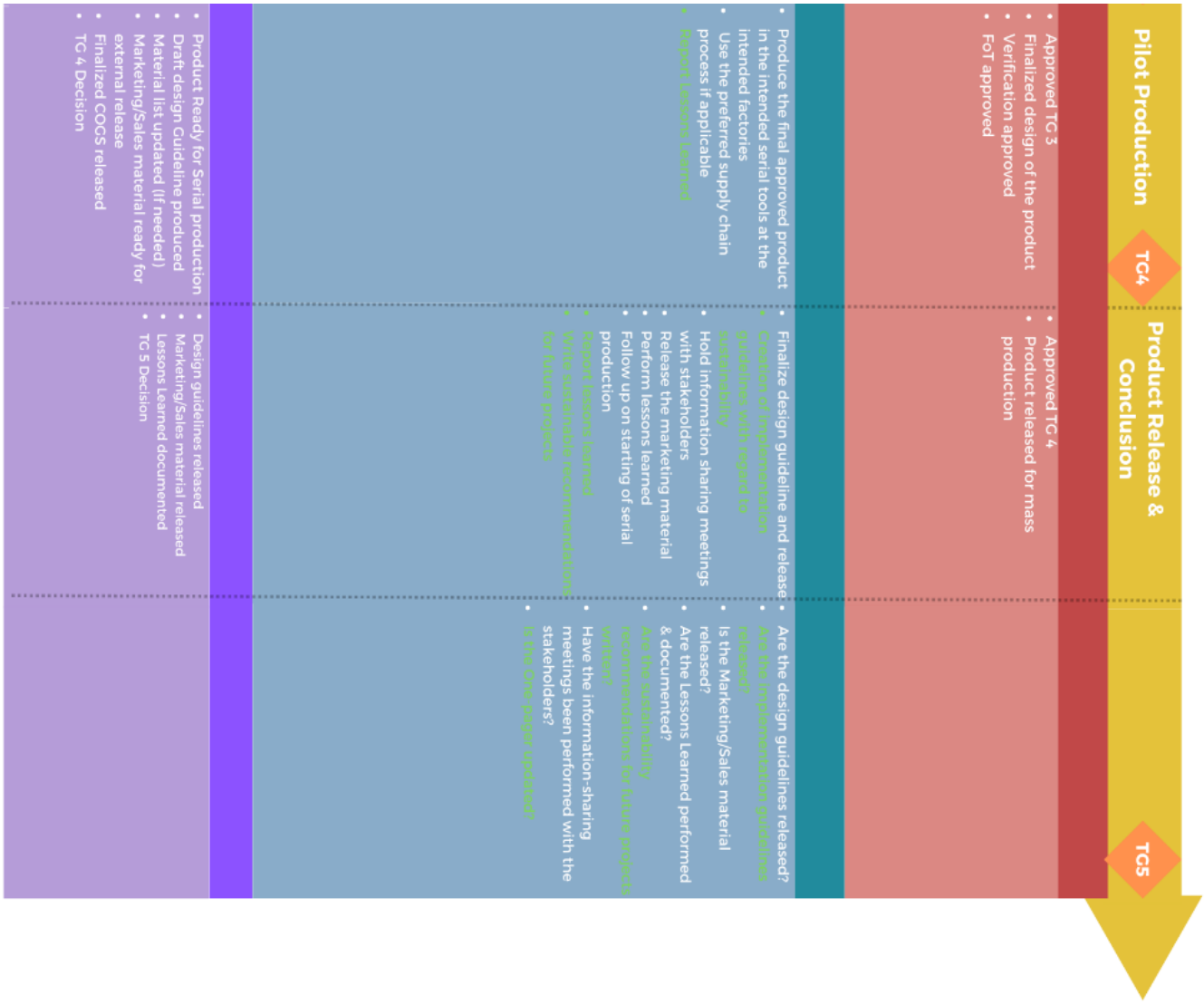


Figure 1. Version 1 of the *Development Process* with changes in green text.

APPENDIX G: SUSTAINABILITY REQUIREMENTS LIST

The *Table of Requirements*, part of the *Requirements list* Excel sheet located in the *Summary File*.

Process	Requirement	Owner	Verified	Type	Example
Development Process	Requirement 6	Product Developer	Yes /No		
	Requirement 5	Product Developer	Yes /No		
	(Updated) Requirement 4	Product Developer	Yes /No		
	Requirement 3	Product Developer	Yes /No		
	(Updated) Requirement 2	Product Developer	Yes /No		
	Requirement 1	Product Developer	Yes /No		
Product Management	Requirement 10	Product Manager	(n/a)	Design for Dissassembly	User is able to dissamble product with household tools
	Requirement 9	Product Manager	(n/a)		Label for recycling // Use modular design for recyclable parts // Use breaking points to separate materials for recycling
	Requirement 8	Product Manager	(n/a)	Design for Recyclability	
	Requirement 7	Product Manager	(n/a)	Product maintenance	Machine washable product in 30 degrees with towels // Weather resistant
	Requirement 6	Product Manager	(n/a)	Product Life	Design for emotional durability // All parts in product shall have same life length
	Requirement 5	Product Manager	(n/a)	Material choice	0% non-recyclable material // Materials shall emit maximum X CO_2 t/ unit
	Requirement 4	Product Manager	(n/a)	Raw material	Use recycled materials // Use recyclable materials
	Requirement 3	Product Manager	(n/a)	Material usage	Padding usage X% lower than Product X
	Requirement 2	Product Manager	(n/a)	Resource consumption during production	Waste X% lower than product X // Produced in a factory with 100% renewable energy // Water usage per unit lower than product X //
	Requirement 1	Product Manager	(n/a)	Toxic substances	Toxic Substance usage X% lower than in product X
Research / Pre-Development	Suggested Requirement 6	RPD Developer	(n/a)		
	Suggested Requirement 5	RPD Developer	(n/a)		
	Suggested Requirement 4	RPD Developer	(n/a)	After-life possibilities	
	Suggested Requirement 3	RPD Developer	(n/a)	Production	
	Suggested Requirement 2	RPD Developer	(n/a)	Materials	
	Suggested Requirement 1	RPD Developer	(n/a)	Emissons	
Innovation Management	Sustainability goals	Innovation Manager	(n/a)	Vague	Contributes to X sustainability goal
	Sustainability potential	Innovation Manager	(n/a)	Vague	Could replace X material with high emissions

Figure 1. Screenshot of the *Sustainability Requirements list template.*

APPENDIX H: FINAL SOLUTION

In Appendix H, a complete summary of the Final Solution is presented in detail as well as all images of the processes. This summary is meant to act as a stand-alone representation of the whole framework and its aim is for the readers or managers to be able to understand the framework without the report.

Abbreviations

CAD	Computer Aided Design
COGS	Cost of Goods Sold
DP	Development Process
FoT	First out of Tool
GHG	Greenhouse gases
IM	Innovation Manager
IMP	Innovation Management Process
IP	Intellectual Property
PM	Product Manager
PMP	Product Management Process
R&D	Research and Development
RPD	Research / Pre-Development
RPDP	Research / Pre-development Process
TG	Toll Gate

The framework presented in this chapter is to be implemented together with the new Development Processes at Mips. The framework is composed of several steps added to these processes as well as six templates that are proposed as a suggestion on how the working documents could look like.

One of the common additions that is to be implemented in all processes is the Sustainability Representative. The decisions taken during the Toll gates are a very important part of the Development Processes. The Decision team is part of all the Toll gates and decides whether the project should be approved and move on to the next phase. To always push the sustainability agenda a Sustainability Representative is recommended to be part of every decision and Toll gate. This individual should be aware of Mips sustainability goals and strategies, best practices for sustainable product development, and have operational knowledge within Mips. Further, during Toll gates a protocol with decisions, reasoning and attendances is should be used for documentation and traceability purposes.

Templates

In this sub-chapter the templates created or modified by the authors and how they are to be used will be presented.

Eco-design rules

These *Eco-design rules* have been adapted and rewritten for Mips. There are two ways that these *Eco-design rules* are meant to be utilized. The first way is to use them as a guide during sustainability studies, requirement definition and concept creations. The second way of using them is to use them as a checklist for decision making and idea/concept evaluation by rephrasing them as questions. Figure 1 presents the rewritten rules to the left and examples of how Mips could implement them in the new product development process to the right.

	Rule	Examples of areas for implementation
1	Reduce or eliminate the use of toxic substances in products	Trace and minimize chemicals in textile production and coating / Study and minimize chemical impact from coatinging of parts
2	Minimise energy and resource comsuption in the production phase and transport through minimise waste, use same parts when possible	Use the same parts for different applications / Design to maximize material efficiency during production (in collaboration with supplier)
3	Minimize weight and material usage in products, if such choices do not interfere with necessary functional priorities.	Minimize padding, part thickness, how much material can be removed before part is no longer functional
4	Consider impact of raw materials (origin, emissions, recyclable, recycled)	Study social sustainability, where does the material come from, is it virgin material? could re-purposed/ recycled material be chosen instead?
5	Minimize use of materials that are hard to recycle or re-purpose or that have a high environmental impact at any point of their life-cycle (including end-of-life).	What part have highest impact? Can those materials be replaced? Considered impact during all life-cycle?
6	Promote long life of products. In implementation, product life should be at least as long as helmet life. Timeless, inclusive design for emotional durability.	Quality materials and durable design.
7	Invest in better materials, surface treatments or structural arrangements to protect products from dirt, corrosion and wear, thereby ensuring reduced maintenance and longer product life	Design for easy maintenance to lengthen product life
8	Prearrange recycling through access ability, labelling, modules, breaking points and manuals.	Maintenance, end of life, recycle instructions / Mips module easy to remove for recycling
9	Promote recycling by using few, simple, recycled, not blended materials and no alloys	
10	Use as few joining elements as possible. Use tools that allow for easy disassembly such as snap fits, geometric locking, adhesives...	Avoid glue, plastic welding

Figure 1. Final *Eco-design rules*.

Idea box template

The *Idea box* template consists of a step-by-step guide on how to add an idea to the Idea Box. This is done by adding a new card containing the new idea. The template located furthest up in the Idea Box. The template is divided in two sections, the first contains the instructions on how to add a new idea, while the second section contains the text that should be pasted on the new card.

The questions in the list ask users to briefly describe their idea, what purpose it has, and what activity it belongs to. The potential of the ideas should also be explained in a few short sentences and not rated as high, medium, or low. This will help point the Innovation Manager in the right direction when further developing the ideas into candidates. The final version of the *Idea box template* is presented in Figure 2.

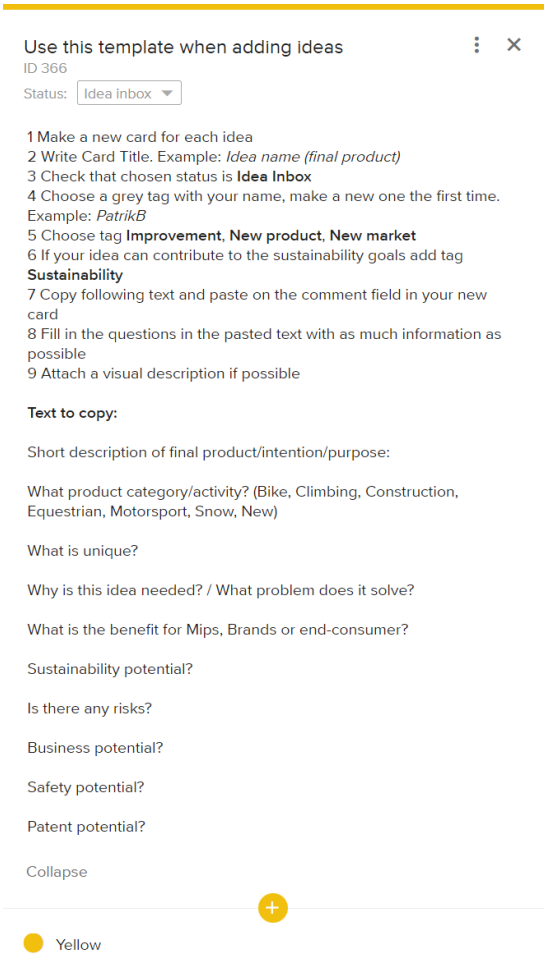


Figure 2. Final version of the *Idea box template*

One-pager

The modified and final version of the *One-pager* is shown in figure x. To better represent and prioritize sustainability needs, a row for sustainability was added to the original template developed by Mips. The idea is that each subprocess has its own sustainability questions that are supposed to be checked and answered at the beginning and end of each process. To give the user a quick overview of the project's progress the most relevant sustainability questions should be documented here. The *One-pager* is the first page of the *Summary file* where other working documents are stored throughout the processes.

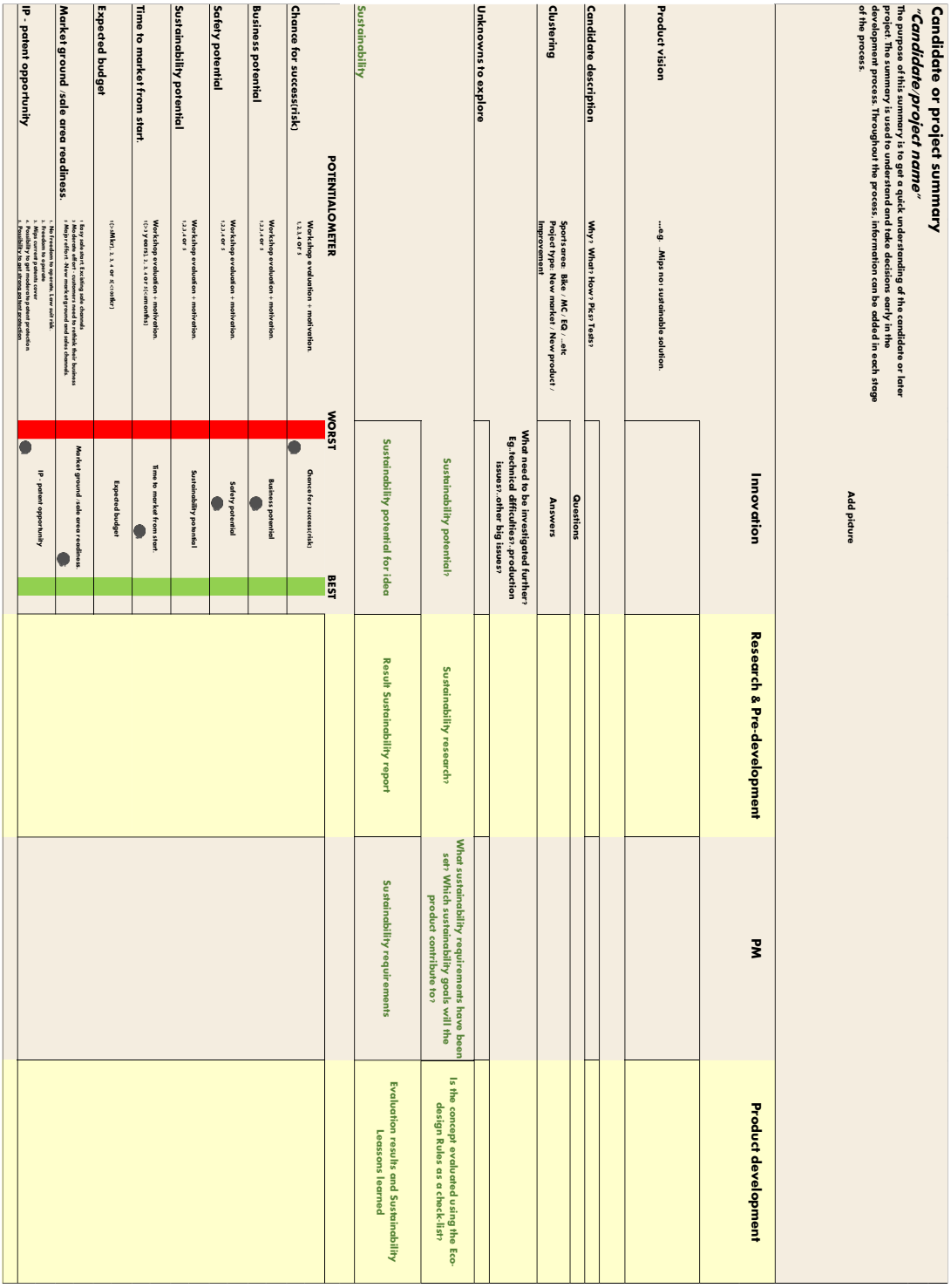


Figure 3. Final version of the *One-pager*.

Sustainability Requirement list

The *Sustainability Requirements list,* Figure 4, is composed of the *Instructions* and the *Requirements table*. The processes are presented in inverse chronological order, with the most current process always on top. Different types of process require different levels of requirements. Therefore, the requirements in the IMP are supposed to be vague, followed by more tangible requirements in the RPDP, and in the PMP the requirements set should be specific and measurable. Therefore, the *Eco-design rules* are to be consulted when setting these requirements. The *Sustainable Requirements list* is the only template that requires the use of the *Eco-design rules* when developing and writing down the *Sustainable Requirements*. Therefore, the *Eco-design Rules* were placed to the side of the *Instructions* section so that they could be consulted when writing the *Sustainable Requirements* during the RPDP and PMP.

Process	Requirement	Owner	Verifie	Type	Example
	Requirement 6	Product Developer	Yes /No		
	Requirement 5	Product Developer	Yes /No		
	(Updated) Requirement 4	Product Developer	Yes /No		
	Requirement 3	Product Developer	Yes /No		
	(Updated) Requirement 2	Product Developer	Yes /No		
Development Process	Requirement 1	Product Developer	Yes /No		
	Requirement 10	Product Manager	(n/a)		
	Requirement 9	Product Manager	(n/a)	Design for Dissassembly	User is able to dissamble product wth household tools
	Requirement 8	Product Manager	(n/a)	Design for Recyclablilty	Label for recycling // Use modular design for recyclable parts // Use breaking points to separate materials for recycling
	Requirement 7	Product Manager	(n/a)	Product maintenance	Machine washable product in 30 degrees with towels // Weather resistant
	Requirement 6	Product Manager	(n/a)	Product Life	Design for emotional durability // All parts in product shall have same life length
	Requirement 5	Product Manager	(n/a)	Material choice	0% non-recyclable material // Materials shall em maximum X CO_2 t/ unit
	Requirement 4	Product Manager	(n/a)	Raw material	Use recycled materials // Use recyclable materials
	Requirement 3	Product Manager	(n/a)	Material usage	Padding usage X% lower than Product X
	Requirement 2	Product Manager	(n/a)	Resource consumption during production	Waste X% lower than product X // Produced in factory with 100% renewable energy // Water usage per unit lower than product X //
Product Management	Requirement 1	Product Manager	(n/a)	Toxic substances	Toxic Substance usage X% lower than in produ X
	Suggested Requirement 6	RPD Developer	(n/a)		
	Suggested Requirement 5	RPD Developer	(n/a)		
	Suggested Requirement 4	RPD Developer	(n/a)	After-life possibilities	
	Suggested Requirement 3	RPD Developer	(n/a)	Production	
	Suggested Requirement 2	RPD Developer	(n/a)	Materials	
Research / Pre-Development	Suggested Requirement 1	RPD Developer	(n/a)	Emissons	
	Sustainability goals	Innovation Manager	(n/a)	Vague	Contributes to X sustainability goal
Innovation Management	Sustainability potential	Innovation Manager	(n/a)	Vague	Could replace X material with high emissions

Instructions

Innovation Management	At this stage the only requiremen is that the sustainability potential The idea should contribute to at least one sustainabiilty goal that
Research / Pre-Development	In this stage the RPD developer will research sustainability aspect Evaluate concept's sustainability l using the Eco-design Rules as a Make suggetions for sustainabilit requirements based on research
Product Management	Evaluate the suggested sustainability requirements Use the Eco-design Rules as a guide to update and define final Incremental innovation products have more specific requirements Radical innovation / New produc will have requirements based on
Development Process	The sustainability requirements a defined in the previous PM They are only to be updated if th need to be changed. The new

Eco-design Rules for Mips

	Rule	Examples of areas for implementation
1	Reduce or eliminate the use of toxic substances in products	Trace and minimize chemicals in textile production and coating / Study and
2	Minimise energy and resource comsuption in the production phase and transport	Use the same parts for different applications / Design to maximize material
3	Minimize weight and material usage in products, if such choices do not interfere	Minimize padding, part thickness, how much material can be removed before part
4	Consider impact of raw materials (origin, emissions, recyclable, recycled)	Study social sustainability, where does the material come from, is it virgin material?
5	Minimize use of materials that are hard to recycle or re-purpose or that have a high	What part have highest impact? Can those materials be replaced? Considered impact
6	Promote long life of products. In implementation, product life should be at	Quality materials and durable design.
7	Invest in better materials, surface treatments or structural arrangements to	Design for easy maintenance to lengthen product life
8	Prearrange recycling through access ability, labelling, modules, breaking points	Maintenance, end of life, recycle instructions / Mips module easy to remove
9	Promote recycling by using few, simple, recycled, not blended materials and no	
10	Use as few joining elements as possible. Use tools that allow for easy disassembly	Avoid glue, plastic welding

Figure 4. Final *Sustainable Requirements list* and instructions.

Design log

The *Design log* is a modification of a file used by one of the process leaders in the past. In this final version a sustainability column was added to keep track of positive and negative results of environmental impact of each iteration of a concept. Therefore, this template is

intended to be used in the development process during the concept and design phase in the RPDP and the DP (see Figure 5).

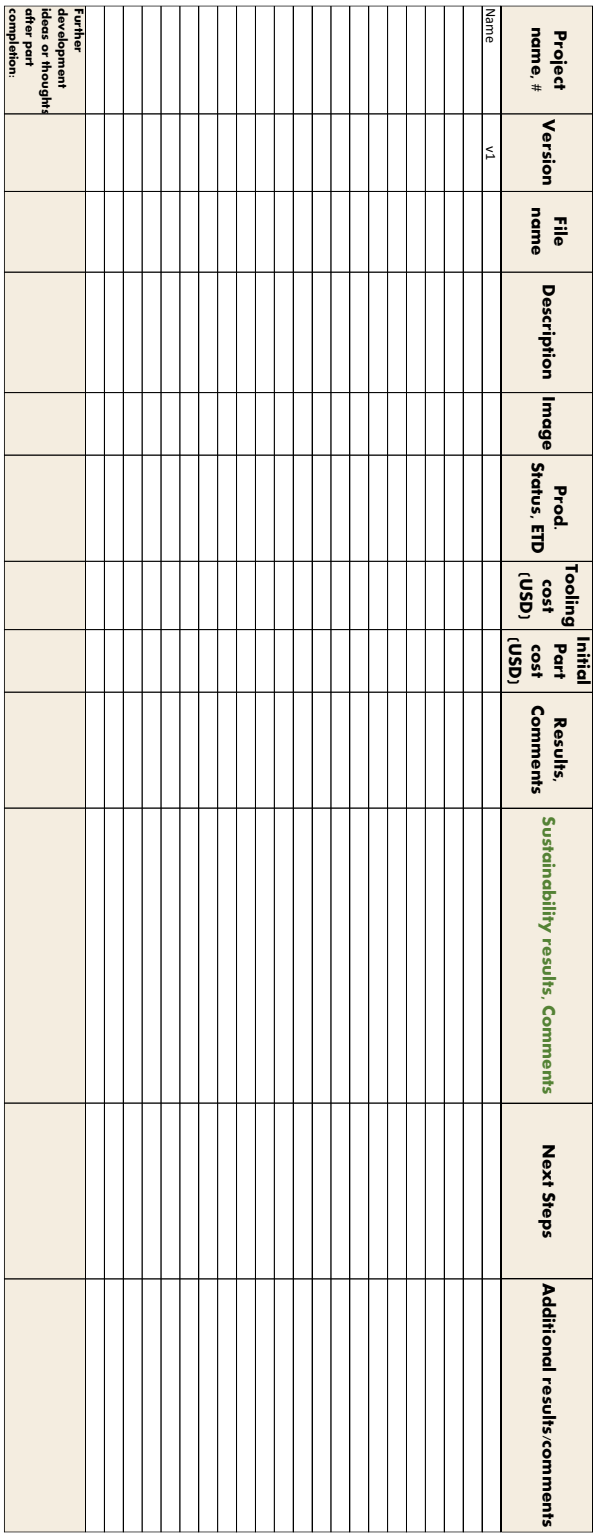


Figure 5. Final version of the *Design log*.

Database template

The Database excel file template is to be used as a working document for the Material Specialists and Operations specialists. Already in the IMP the document can be consulted when defining the sustainability potential for the idea candidates. In the RPDP this document can be both consulted during the process and later updated with the sustainability findings at the end of the phase. During the PMP the document could be consulted while defining the requirements. Finally, during the DP this document can be consulted when making decision regarding the concept, materials and production techniques and later updated after the project's conclusion with any learnings on the materials or sustainability aspects.

Fabric and foam

The first sheet, called *Fabric and foam*, is for the Textile specialist to fill in data available on the different materials. This template (see Figure 6) was developed by the Textile Specialist.

Mi (Mips), K (Knitted), W (Woven), N (Non woven), F (Foam), I (insulation) W (Woven)							*Stretch properties:*		Link to test					Solution dye / recycled / oeko-tex dye / Sorons fiber						
Type	**Art. Number**	**Fabric name**	**Replace**	**Weight / Thickness**	**Fabric Width**	**Col.ways**	**Artical no. Supplier**	**Supplier**	**Composition**	**OLD replace**	**Finish**	**Test reports / Data sheet**	**Washing instruction**	**Properties Marking**	**Sustainable Marking**	**MOQ**	**Price(RMB/ Yard)**	**Notes**	**Marking**
Knitted	XX X XX	Name		140g/m2	62"	Black Dark grey	XXXXXX	XX	*89% RECYCLED POLYESTER 12% ELASTANE*	*90% Polyester* *10% Spandex (Elastane)*	N/a	Link	Machine wash 30°C Do not bleach Mild drying processes Iron at low temperature Do not dry-clean	Moisture-Wicking Recycle polyester	Oeko-Tex Blue sign	Min Qty / order 3000 yard. Min Qty/ color 1000 yards	3,05US/Yards		

Figure 6. *Fabric and Foam* template

Materials (polymer)

The second sheet is the material sheet. The first part of the table is for the Material specialist to fill in the general and supplier information given by the suppliers. This template (see Figure 7, 8 and 9) was developed by the Material Specialist.

Base material	Application	Suppliers	Product name from supplier

Figure 7. General material and supplier information.

Second part of the table regards the mechanical properties of the material (see Figure 8).

Tensile stress at break (MPa)	Tensile strain at break (%)	Tensile stress at yeild (MPa)	Tensile strain at yeild (%)	Tensile modulus (MPa)	Flexural Strength (MPa)	Flexural modulus (MPa)

Notched impact strength 23°C (kJ/m2)	Notched impact strength 20 °C (kJ/m2)	Heat Deflection Temperature (°C)	Water absorption 23°C, 50%RH (%)	Mold Shrinkage 3T, MD/TD (%)	Other properties

Figure 8. Mechanical properties of the material.

The last part regards the testing results (see figure 9).

Testing result								
Fatigue test	Cold Impact	Temperature variation test	High temp. high humidity	Artificial aging test	Chemical tests	MIPS function test	Other tests	Link to test reports

Figure 9. Testing results.

Suppliers

The third sheet, called *Suppliers* (see Figure 9), is meant to be filled by the Operations specialist with information regarding basic information regarding the suppliers such as name, country, manufacturing techniques, if the use green energy, etc.

Name	Country	Manufacturing techniques	Green Energy	Certifications	Transparency	Products	Latest	Comments
Green Components	Germany	Plastic molding	Yes	Certified Carbon Neutral, Energy Star machinery	100% - energy consuption, water usage, ceritified green energy	All plastic components for Integra X	15 April 2023 - Start date for manufacturing set	Change all other products to this supplier?

Figure 9. *Suppliers* template.

New technology

The fourth and final sheet is meant to document new technologies regarding different areas (see Figure 10) such as manufacturing, design, etc. Furthermore, the potential of these technologies is to be documented especially if it can help Mips become even more sustainable. This sheet is meant to be used by anyone that is part of the R&D department.

Area	Summary	Project	Potential	Tests	Results	Latest	Next step	Owner
Manufacturing	Researchers at X have found new 3D printing technique	Project 123	Less material usage in Mips products, new more complex shapes can be used			15 April 2023 - Test machine ordered.		Researcher 1
Design	Sustainability finding X	Project 584	Concepts using X Design are much more sustainable because of Y and Z			Project started to check current products and re-asses design	Start re-designing process of product X, Y?	Developer 3

Figure 10. *New technology* template.

Product development process

In this sub-chapter, all the sustainability steps added to each subprocess will be presented.

Innovation Management Process

To promote sustainability thinking as early as possible, sustainability topics should be used when ideas are generated. Therefore, some sustainability topics are suggested to be used during idea generation. These topics are, for example, circularity, direct input from customers/helmet brands, input from other departments, radical/incremental innovation, products lifecycle, or end of life. A step-by-step guide on how to generate and add ideas to the *Idea box* is presented earlier. In the Second Selection phase the Innovation Manager should consult the *Database* template to evaluate the potential of the candidates and update their *One-pager*. Further, the Innovation Manager is to set vague sustainability requirements for the selected *candidates* by using the *Sustainability Requirement list.* See Figure 11 the added steps in green.

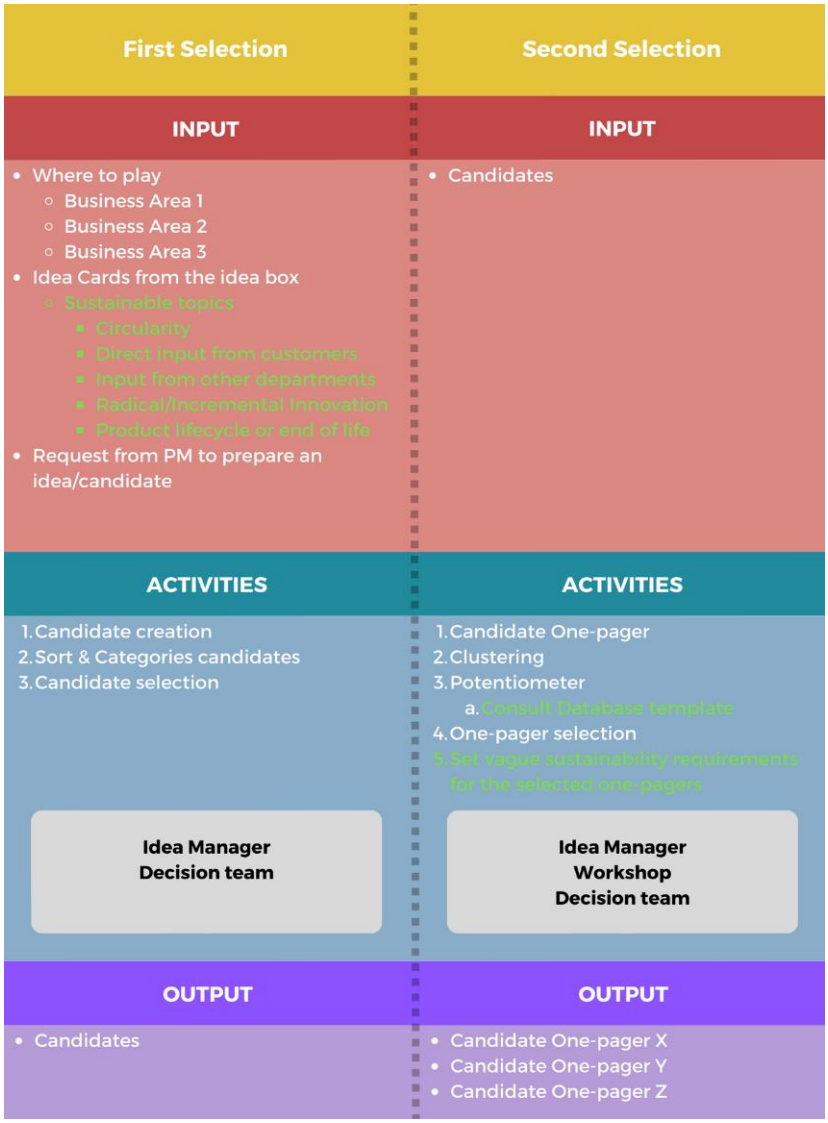


Figure 11. Final version of the *Innovation Management process*.

Research and Pre-development Process

In *Idea handover* phase a *concept/product brief* was added as an activity. The brief is to include information regarding sustainability and a *One-pager*. Additionally, a Sustainability Representative is to be present during this brief and be a part of the decision to determine the fate of the project (see Figure 12).

During the *Initial research* phase, the RDP developer is to use the *Eco-design rules* to assess the current idea. Further, the developer is to conduct sustainability research to check if the *Eco-design rules* can be applied, if there are alternatives of materials, possibilities of green production, after-life possibilities, and emissions emitted for products of similar type or size. Additionally, the RDP developer is to start documenting findings in a report.

In the *Pre-development* phase, the RDP developer is document sustainability implications of the concept that is created, tested, and prototyped in the *Design log.* The RPD developer is also to continue with the documentation of sustainability findings.

In the *Concept finalization* phase, the RDP developer is to make suggestions for the product sustainability requirements, update the *One-pager* and the potentiometer in it, and lastly continue with documentation of any sustainability findings during this phase. The RDP developer is to use the *Sustainability Requirements list* to create the suggestions and document it in the list as well.

Finally, in the *Concept handover* phase the RDP developer is to hand over an RDP report that includes the result of the sustainability research, an updated *One-pager,* and a list of suggestions of potential *Sustainability requirements.* During the final gate the participants are to discuss the result of the assessment. Besides the concepts pro & cons, risk, market segment etc, the participants are to discuss the findings of the *Sustainability Research* and how it could be applied in the upcoming process.

Figure 12. Final version of the *Research & Pre-development Process.*

Product Management Process

To start, the *Insights* phase, the candidate *One-pager* from the Innovation Management process should contain a sustainability potential and the concept's brief from the Research & Pre-development should include suggestions of possible sustainability requirements. During the evaluation of the concepts and candidates the Product Manager should consider the sustainability wishes of the end users, if a sustainability business model is applicable, if the candidate/concepts will help Mips reach their sustainability goals, and if the sustainability roadmap and product roadmap are aligned with each other. After the *Insight phase* measurable requirements for the product should be set by the Product Manager and a Sustainability Manager or Expert and the requirements should be set by using the *Sustainability Requirements list* template. If the selected concept has been through the Research & Pre-development process, the suggestions on possible requirements should be considered. See Figure 13.

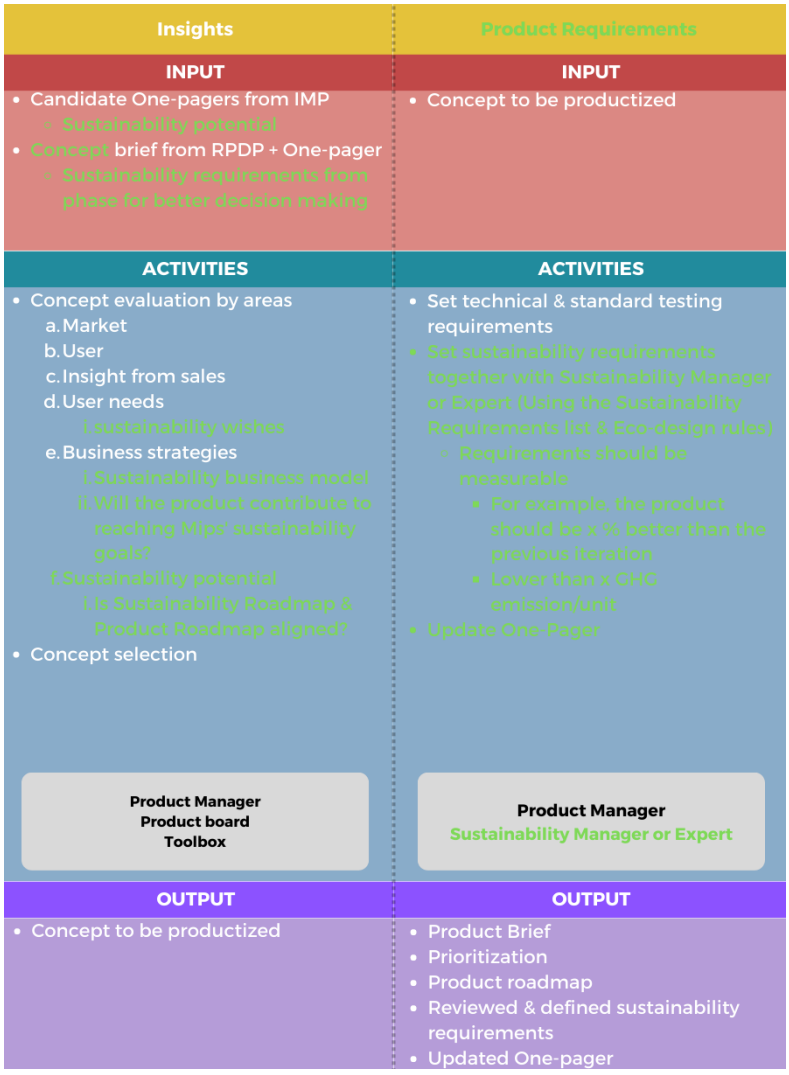


Figure 13. Final version of the *Product Management process.*

In the *Toolbox* used during this phase, several tools were added (see Figure 14). In *Market Insight* the customer's sustainability views are to be taken into consideration, in *Brand Values* and *Business Plan* sustainability were added as a factor to study, and finally a whole new tool was added called *Sustainability Roadmap*. This last tool should be aligned with the *Product Roadmap* so that a clear correlation is shown between the sustainability goals

and milestones and to make sure that they are aligned with the products planed for release, as well as the products that need to be phased out.

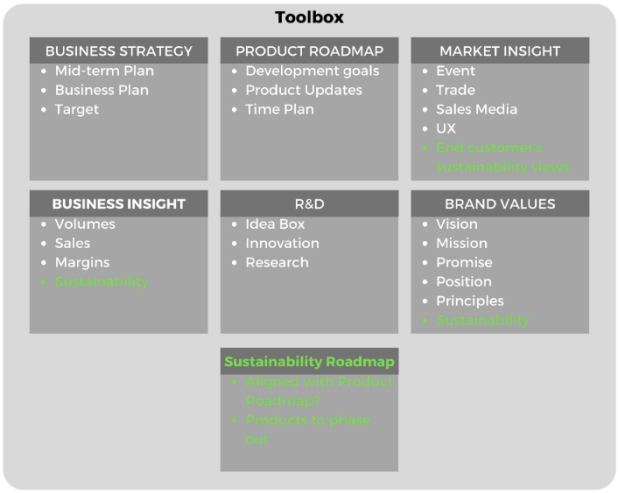
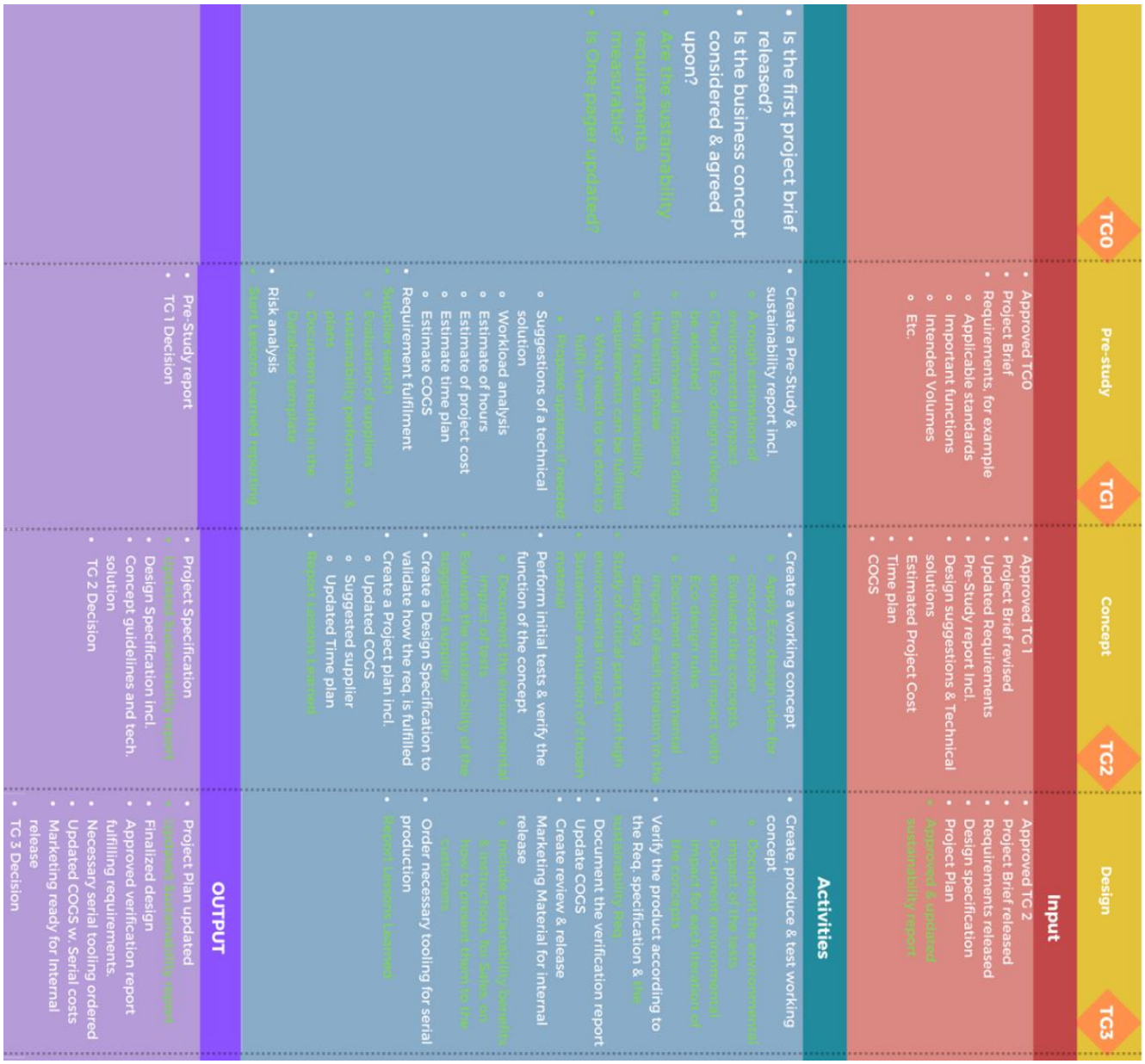


Figure 14. Final version of the *Toolbox* used during the PMP.

Development Process

In this subchapter all the steps added to the Development Process will be presented. An overview of changes can be found in Figure 15.



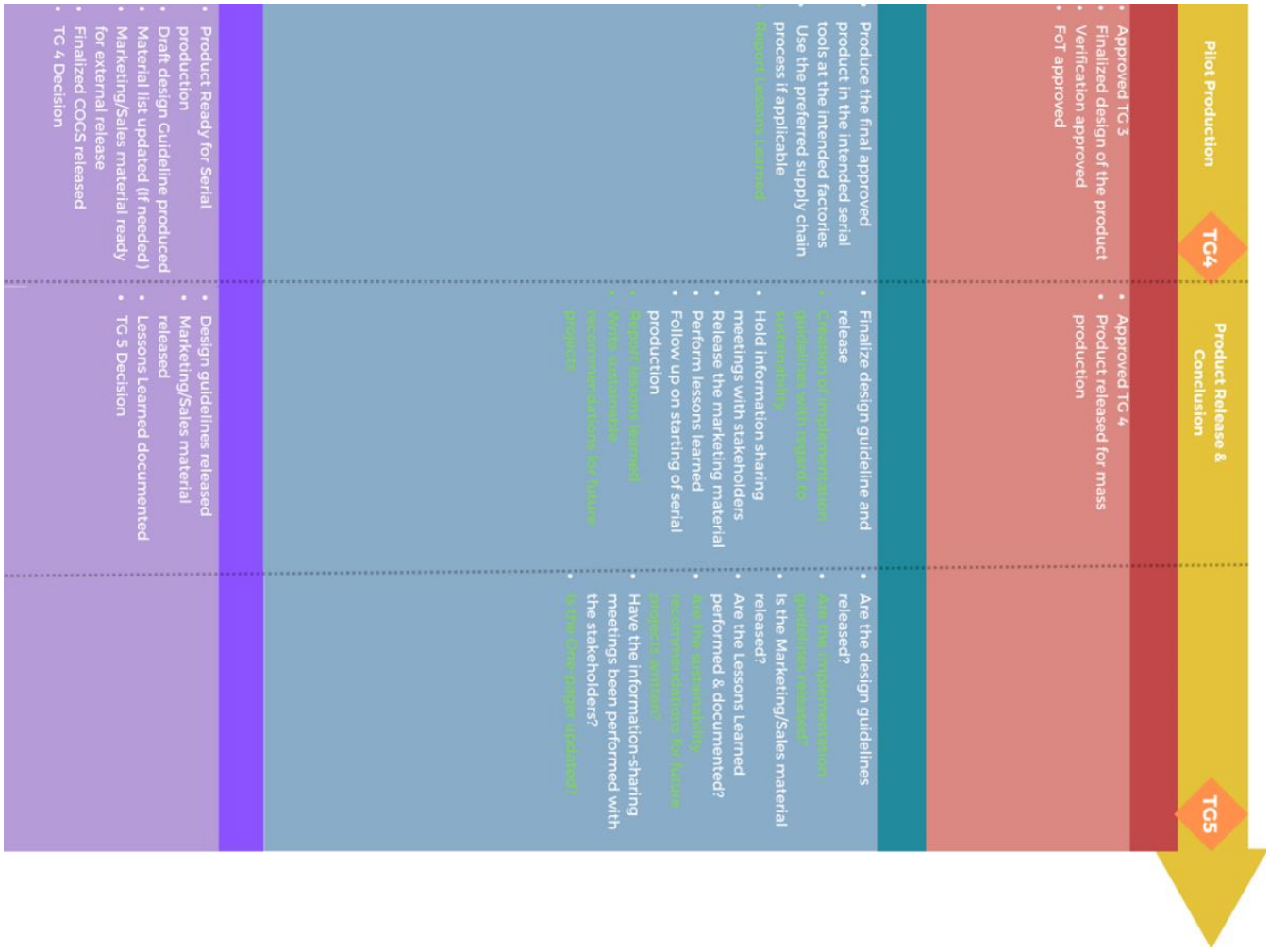


Figure 15. Overview of steps added to the *Development Process*.

Toll Gate 0

Since the product requirements are meant to be set in the PMP there needs to be a discussion point in TG0 about the sustainability requirements to check whether they have been set and if they are measurable. Another discussion point is to check if the *One-pager* have been updated (see Figure 16).



Figure 16. Discussions points in green text.

Pre-study

In *Pre-study* phase a sustainability study should be performed. The study is to find out if the set sustainability requirements are verifiable and how to verify them. If they cannot be verified the requirements should be updated. Another goal of the study is to make a rough estimation of the greenhouse gases emitted to create a product of similar size or kind. Moreover, an estimation of emissions for the upcoming tests should be performed.

Furthermore, the study is also to check if the *Eco-design rules* can be adapted for the project. Additionally, during the supplier search a simple evaluation of the supplier's sustainability performance and plans should be performed and the suppliers' data gathered and stored in the *Database Template*. Findings of the study should be documented in a report that is to be presented in the upcoming gate. The developer/researcher should also create and start a documentation of lessons learned which is to be continuously updated during the whole DP (see Figure 17).



Figure 17. Sustainability tasks for the *Pre-study* phase in green.

Concept

In the *Concept* phase the *Eco-design rules* are be taken in consideration when creating the concepts and when evaluating them. Further, a list of parts that are most critical should be created and the list should contain emission data of each part. Moreover, a sustainable evaluation of the chosen materials should also be performed to confirm that they fulfil the *Sustainability requirements*. During the initial testing of the concepts, the iterations and how changes made will impact the concepts sustainability impact should are to be documented in the *Design log*. Furthermore, the environmental impact of the initial testing is to be recorded. Additionally, the intended suppliers should also be evaluated at this point (see Figure 18).

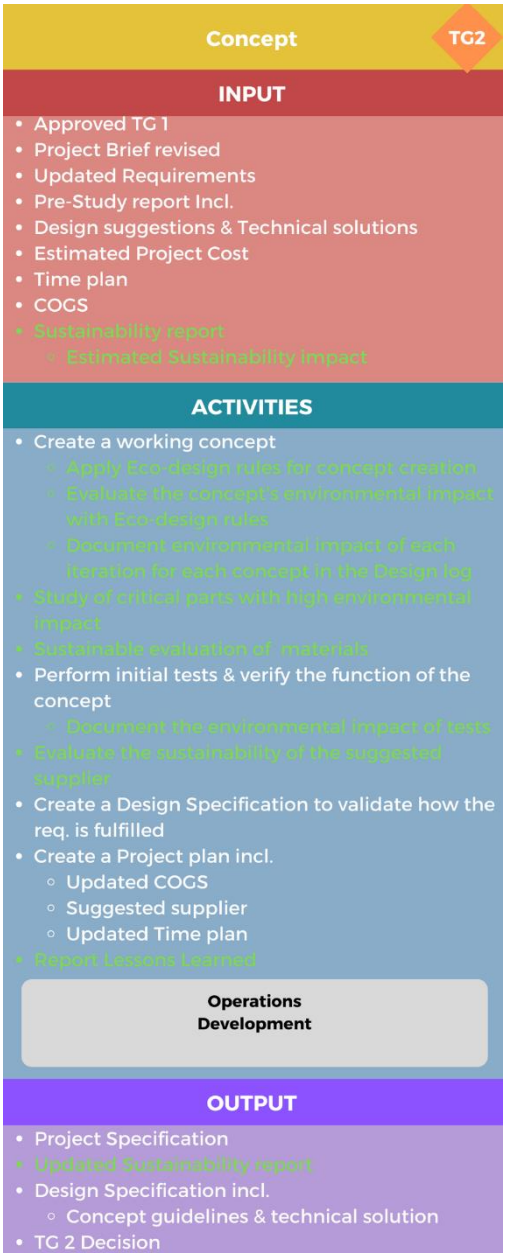


Figure 18. Sustainability tasks for the *Concept* phase.

Design

In the *Design* phase the documented iterations of concepts should be continued in the *Design log*. Moreover, during the verification of product requirements, the sustainability

requirements should also be verified. When the marketing material is to be created it should contain the product's sustainability benefits and instructions that will allow the Sales department to easily present said benefits to the customers (see Figure 19).

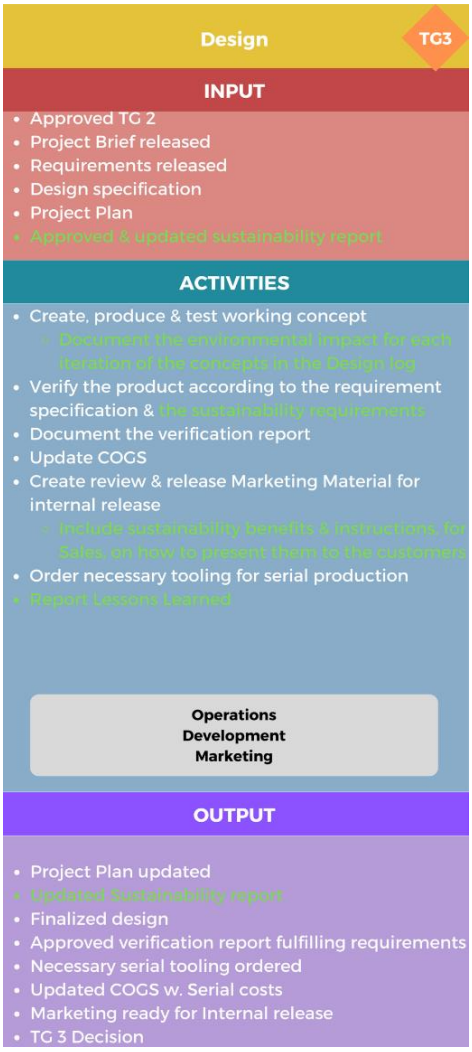


Figure 19. Sustainability tasks for the *Design* phase in green text.

Pilot Production

In the *Pilot Production* phase, it is deemed that any changes to make the product more sustainable are too late be implemented. Therefore, the only added task to do in this phase is to document any lessons learned that can improve the process in the future (see Figure 20).

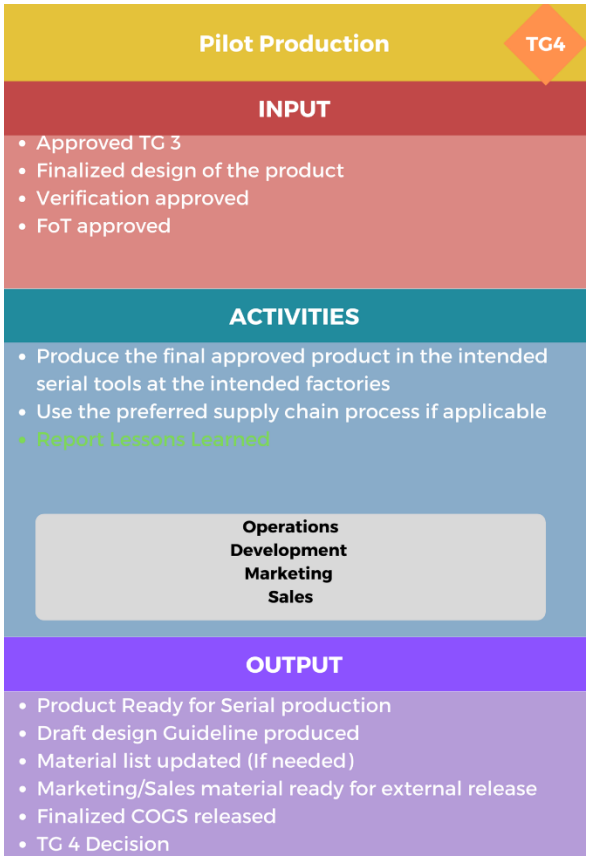


Figure 20. Sustainability tasks for the *Pilot Production* phase.

Product Conclusion & Release

In the last phase, *Product Release & Conclusion*, the implementation guidelines should be created together by stakeholders from R&D and implementation. Further, lessons learned during the whole project is to be written into recommendations for future projects (see Figure 21).



Figure 21. Sustainability task for the *Product Release & Conclusion* in green.

Toll Gate 5

Lastly, during *Toll gate five* the participants are to check if the implementation guidelines have been released, if final updates have been done to the *One-pager*, and if the recommendations for future projects have been written. See Figure 22 for the tasks added to the process.

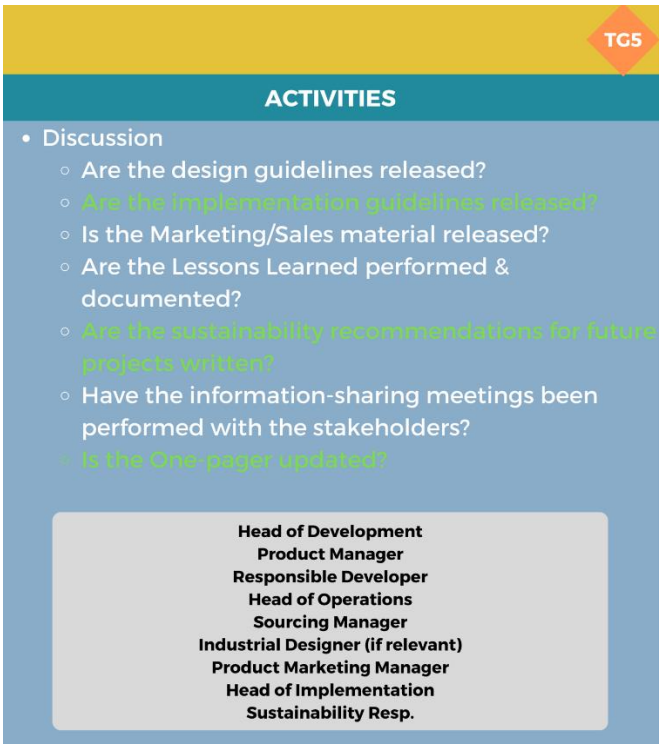


Figure 22. Sustainability discussion points in green for *Toll Gate 5.*

APPENDIX I. DATABASE TEMPLATE

A suggestion for a *Database* file that could be used to start gathering information on regarding materials, suppliers, new materials and technologies, fabrication methods, and important sustainability findings.

Fabric and foam

An example of how the file could look like for fabric and foams developed by the textile specialist.

Mi (Mips), K (Knitted), W (Woven), N (Non woven), F (Foam), I (insulation)
W (Woven)

Stretch properties: Link to test

Solution dye / recycled / oeko-tex dye / Sorons fiber ....

Type	Art. Number	Fabric name	Replace	Weight / Thickness	Fabric Width	Col.ways	Artical no. Supplier	Supplier	Composition	QLD replace	Finish	Test reports / Data sheet	Washing instruction	Properties Marking	Sustainable Marking	MOQ	Price(RMB/ Yard)	Notes	Marking
Knitted	XX X XX	Name		140g/m2	62"	Black Dark grey	XXXXXX	XX	*88% RECYCLED POLYESTER 12% ELASTANE*	90% Polyester 10% Spandex (Elastane)	N/a	Link	Machine wash 30°C Do not bleach Mild drying processes Iron at low temperature Do not dry-clean	Moisture-Wicking Recycle polyester	Oeko-Tex Blue sign	Min Qty / order 3000 yard. Min Qty/ color 1000 yards	3,05US/Yards		

Figure 1. Fabric and Foam template

Materials (polymer)

An example of how the file could look like for a polymer template developed by the material specialist. First part of the table regards general material and supplier information.

Base material	Application	Suppliers	Product name from supplier

Figure 2. General material and supplier information.

Second part of the table regards the mechanical properties of the material.

Tensile stress at break (MPa)	Tensile strain at break (%)	Tensile stress at yeild (MPa)	Tensile strain at yeild (%)	Tensile modulus (MPa)	Flexural Strength (MPa)	Flexural modulus (MPa)

Notched impact strength 23°C (kJ/m2)	Notched impact strength 20 °C (kJ/m2)	Heat Deflection Temperature (°C)	Water absorption 23°C, 50%RH (%)	Mold Shrinkage 3T, MD/TD (%)	Other properties

Figure 3. Mechanical properties of the material.

The last part regards the testing results.

Testing result								
Fatigue test	Cold Impact	Temperature variation test	High temp. high humidity	Artificial aging test	Chemical tests	MIPS function test	Other tests	Link to test reports

Figure 4. Testing results.

Suppliers

An example of how the file could look like for supplier information.

Name	Country	Manufacturing techniques	Green Energy	Certifications	Transparency	Products	Latest	Comments
Green Components	Germany	Plastic molding	Yes	Certified Carbon Neutral, Energy Star machinery	100% - energy consuption, water usage, ceritified green energy	All plastic components for Integra X	15 April 2023 - Start date for manufacturing set	Change all other products to this supplier?

Figure 5. Supplier information template.

New technology

An example of how the file could look like for new technology.

Area	Summary	Project	Potential	Tests	Results	Latest	Next step	Owner
Manufacturing	Researchers at X have found new 3D printing technique	Project 123	Less material usage in Mips products, new more complex shapes can be used			15 April 2023 - Test machine ordered.		Researcher 1
Design	Sustainability finding X	Project 584	Concepts using X Design are much more sustainable because of Y and Z			Project started to check current products and re-asses design	Start re-designing process of product X, Y?	Developer 3

Figure 6. New technology template.

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